Lesson 1 — Statistical Modeling

Random Variables, Distributions, LLN, CLT

Applied Statistics Course

September 16, 2025

Outline

1. Modeling Motivation
2. Foundations and Random Variables
3. Statistics and Convergence
4. Limit Theorems
5. Exercises
6. Summary and References

Why Probability? From Questions to Models

Natural language $\rightarrow$ **mathematical model** via events and random variables.

- Web A/B test: "Is variant B better than A?" $\Rightarrow$ Encode clicks as Bernoulli trials; compare p_A vs p_B.
- Manufacturing: "Are defects random and rare?" $\Rightarrow$ Model counts with Poisson; check fit.

Language of sets and probability

Events are sets (subsets of Ω). Questions like "did a click occur?" or "defects ≤ 3" translate to $A \in \mathcal{F}$ and $\mathbb{P}(A)$.

Probability theory provides mathematical tools designed to **formalize uncertainty** and enable rigorous **reasoning about random phenomena**.

From Natural Language to Events and RVs

Natural question	**Event / RV**	**Typical model**
"Roll of a fair die?"	$X \in \{1, \ldots, 6\}$, event $E = \{X \text{ even}\}$	X uniform on $\{1, \ldots, 6\}$; $\mathbb{P}(X = k) = 1/6$
"User clicks?"	$C = \{\text{click}\}$, indicator $X = \mathbf{1}_C$	$X \sim \text{Bernoulli}(p)$; compare p_A vs p_B
"Defects in a batch?"	Count $D \in \{0, 1, 2, \ldots\}$	$D \sim \text{Poisson}(\lambda)$ (rare, independent)
"Time until failure?"	Continuous $T \geq 0$	$T \sim \text{Exponential}(\lambda)$ (memoryless)

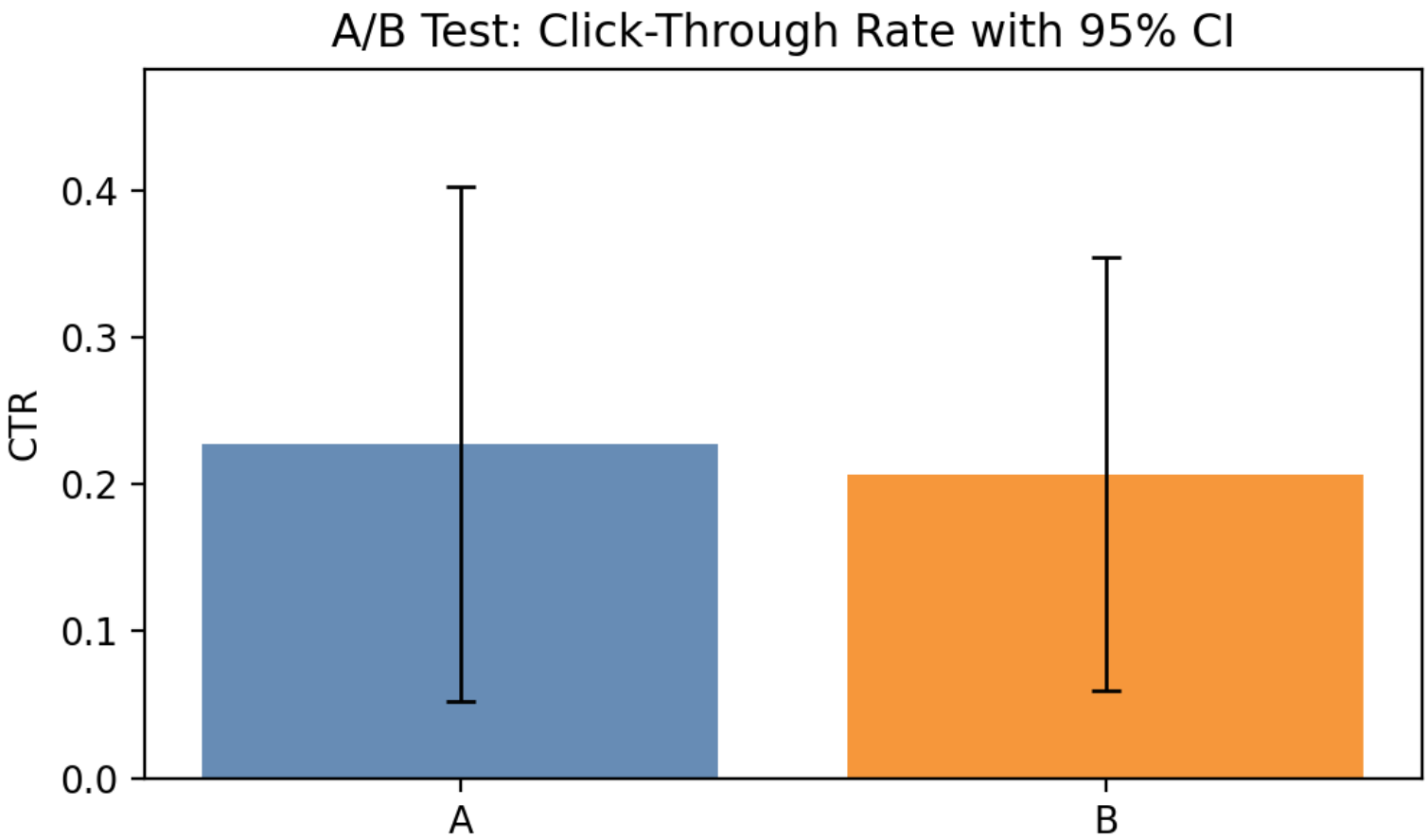
Set operations encode logic:

$A \cup B$ = "A or B", $A \cap B$ = "A and B"

A^c = "not A"

Example: A/B Test as Bernoulli/Binomial

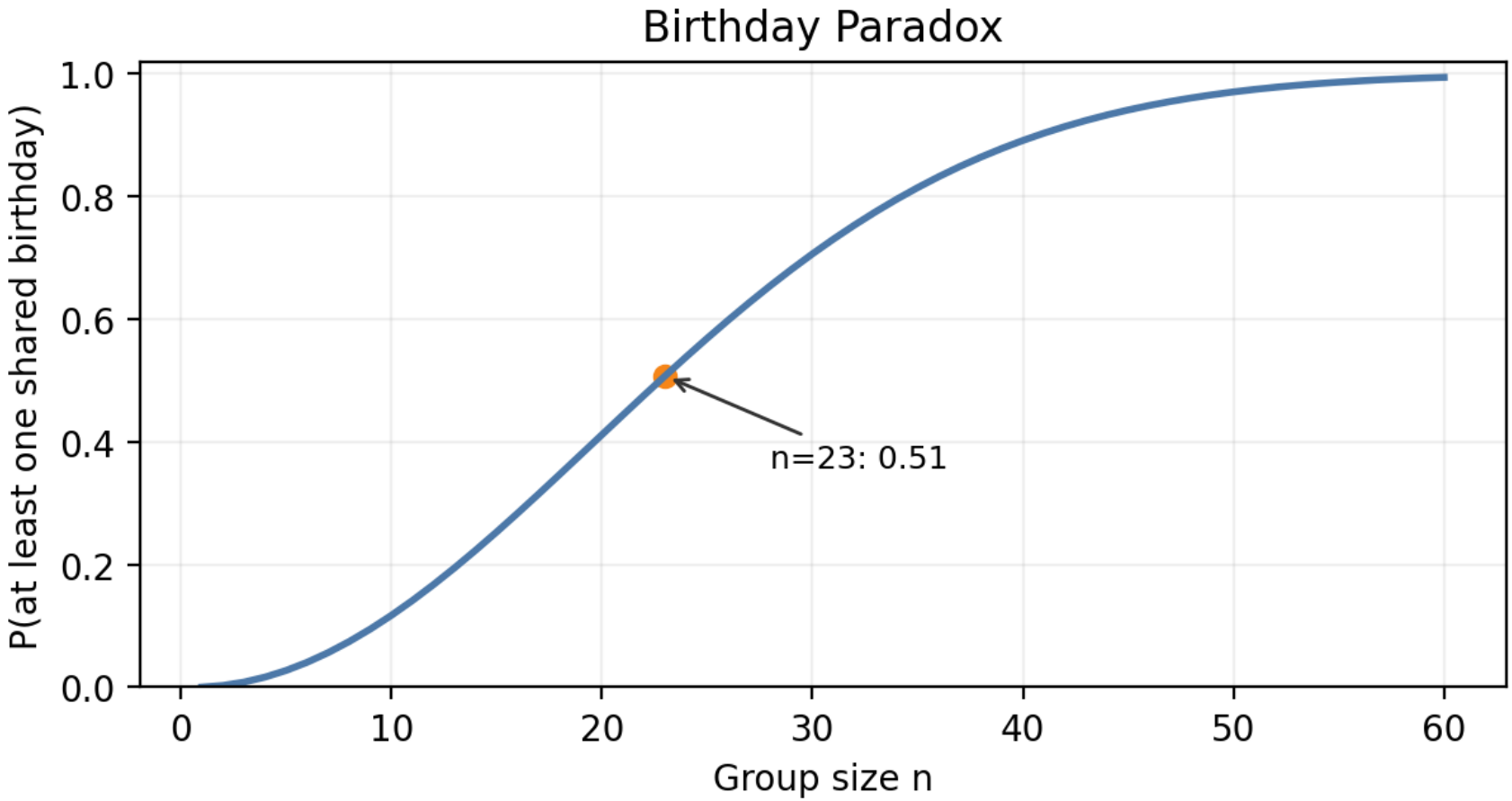
Each impression is a trial: click $= 1$, no click $= 0$. Variant-level CTR estimates $\hat{p} = \frac{\text{clicks}}{\text{impressions}}$.



Confidence intervals visualize uncertainty from finite samples.

Catchy: The Birthday Paradox

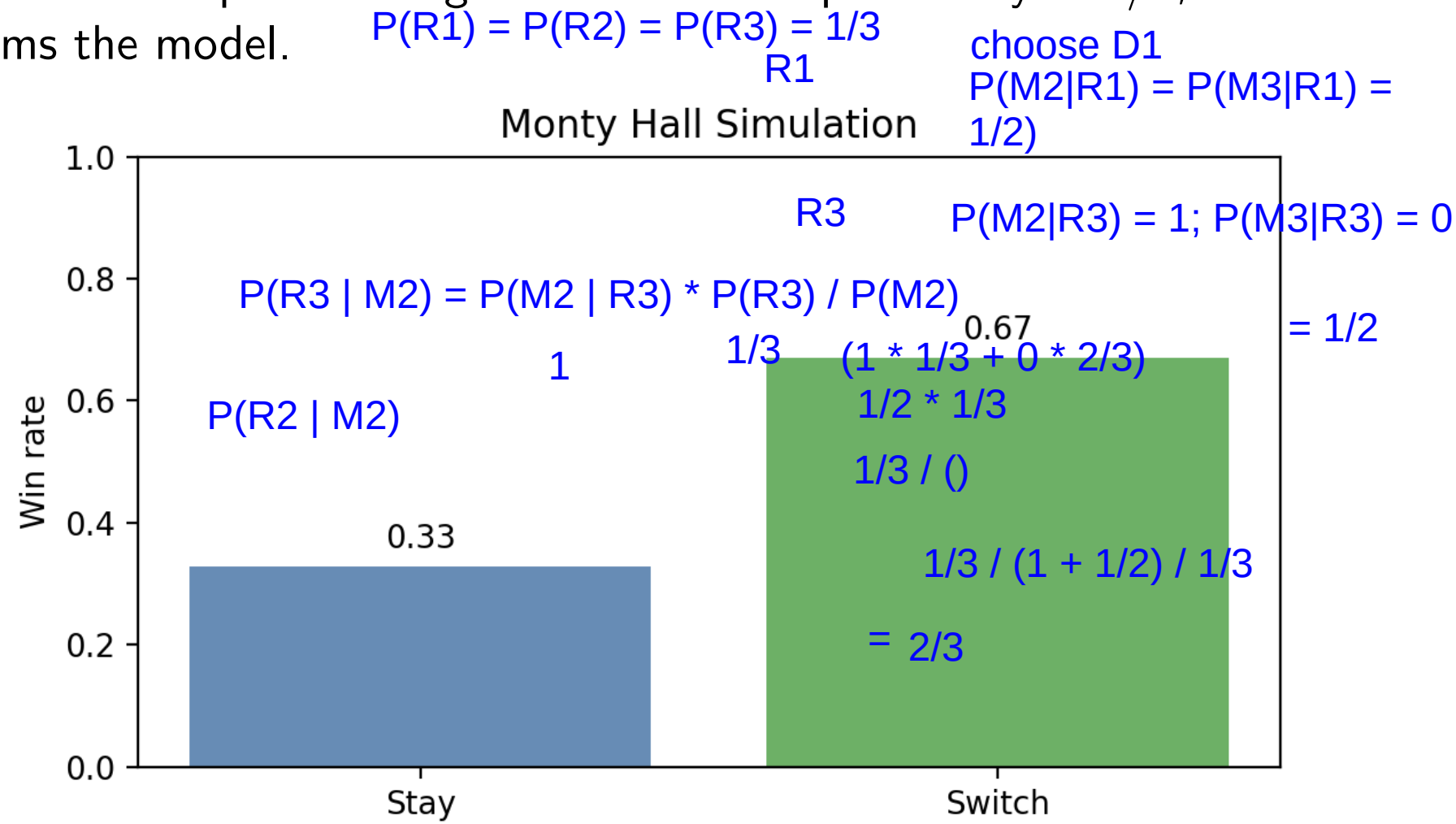
In a class of n students, what's the chance at least two share a birthday? Surprisingly, at $n = 23$ it's about 0.5.



Shows how multiplicative complements and event counting yield unintuitive results.

Catchy: Monty Hall — Switch or Stay?

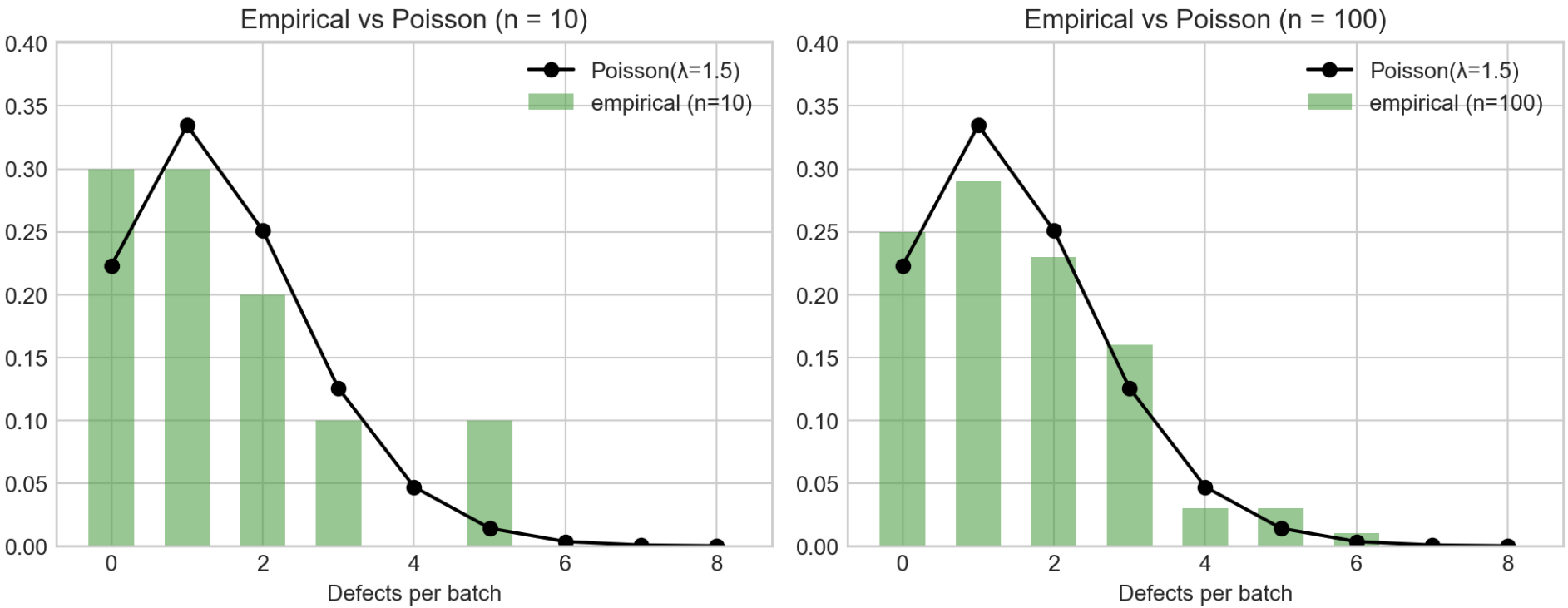
Game show setup: switching doors wins with probability $\approx 2/3$; simulation confirms the model.



Encodes information and conditional probability; a great motivator for Bayes' rule.

Example: Defects as Poisson Counts

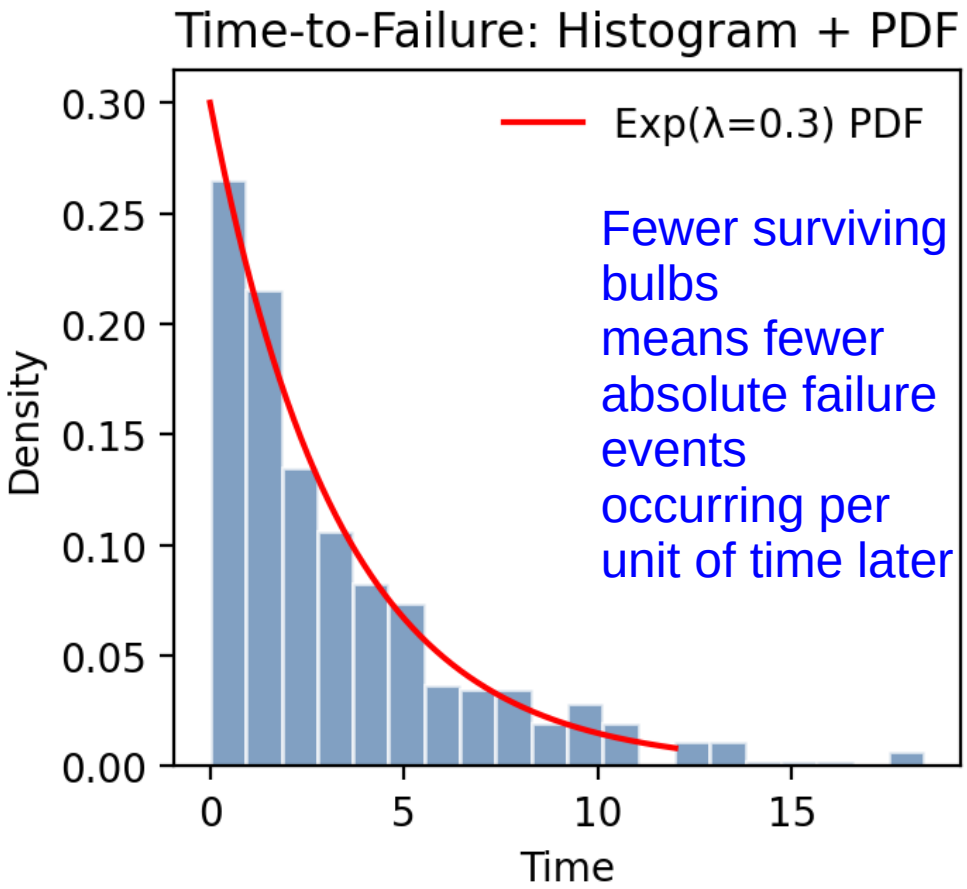
When events are rare and independent, counts in a fixed window often follow Poisson(λ). Compare empirical distribution vs Poisson model.

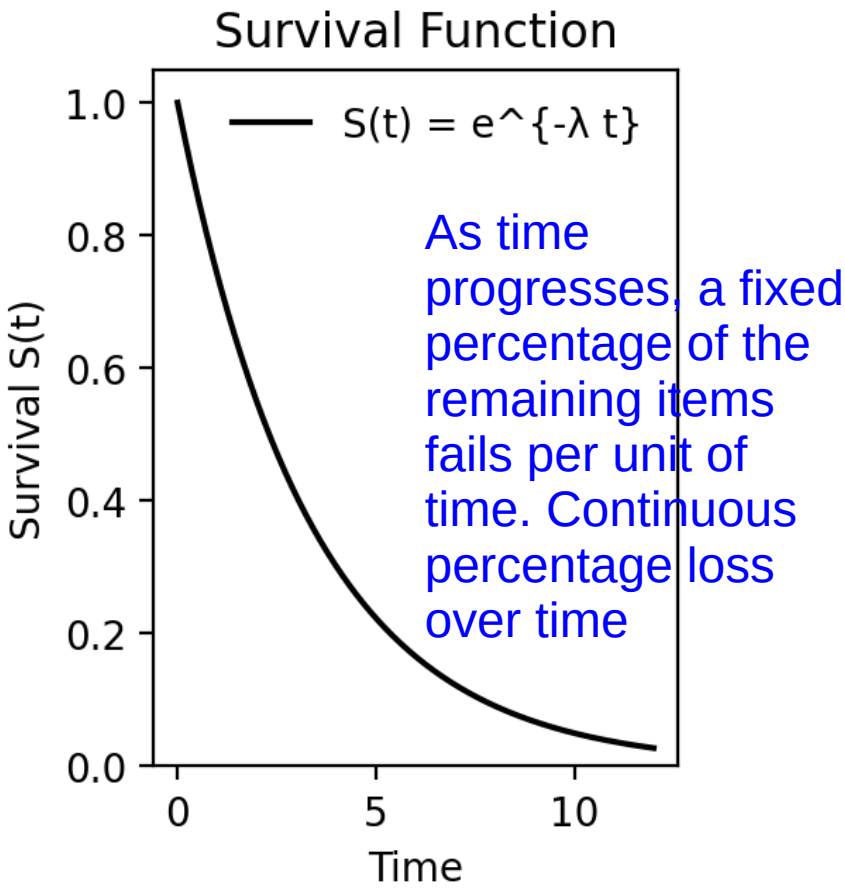


If the fit is reasonable, λ summarizes the average defects per batch, guiding quality control.

Example: Time-to-Failure as Exponential

Waiting times between independent events are often modeled as Exponential(λ). The survival is $S(t) = \mathbb{P}(T > t) = e^{-\lambda t}$.





Useful for reliability, queueing, and risk modeling.

Learning Objectives

- **Define random variables (RVs) and distributions.**
  Purpose: Map real-world uncertainty to mathematical objects and clarify what a distribution encodes.
- **Use PMF/PDF/CDF to compute probabilities/quantiles.**
  Purpose: Turn event and threshold questions into computations to answer "how likely?" and "what cutoff?".
- **Compute expectation, variance, and interpret moments.**
  Purpose: Summarize center and variability to compare models and quantify uncertainty/risk.
- **Understand LLN and CLT (intuition + statements).**
  Purpose: Explain why averages stabilize and when normal approximations apply, enabling CIs and tests.
- **Connect descriptive statistics to probabilistic modeling.**
  Purpose: Bridge EDA to formal models so assumptions are explicit and limits understood.

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Probability Space – Recap

Definition. A probability space is a triple $(\Omega, \mathcal{F}, \mathbb{P})$ where:

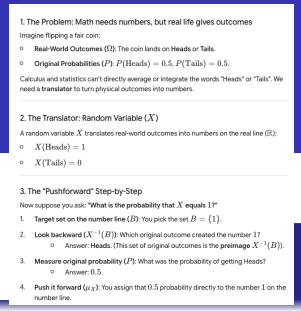
- Ω is the sample space; $\mathcal{F} \subseteq 2^{\Omega}$ is a σ-algebra;
- $\mathbb{P} : \mathcal{F} \to [0,1]$ is a probability measure, $\mathbb{P}(\Omega) = 1$, and $\mathbb{P}$ is countably additive.

Basic properties for events $A, B \in \mathcal{F}$

- Bounds: $0 \leq \mathbb{P}(A) \leq 1$, with $\mathbb{P}(\varnothing) = 0$ and $\mathbb{P}(\Omega) = 1$.
- Complement: $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
- Monotonicity: $A \subseteq B \;\Rightarrow\; \mathbb{P}(A) \leq \mathbb{P}(B)$.
- Union/intersection: $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.
- Disjoint additivity: if $A \cap B = \varnothing$, then $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$; extends to countable disjoint unions.

Events are the measurable statements about outcomes: elements of $\mathcal{F}$.

Random Variables and Laws


1. The Problem: Math needs numbers, but real life gives outcomes

Imagine flipping a fair coin:

- **Real-World Outcomes (Ω):** The coin lands on **Heads** or **Tails**.
- **Original Probabilities (P):** $P(\text{Heads}) = 0.5$, $P(\text{Tails}) = 0.5$.

Calculus and statistics can't directly average or integrate the words "Heads" or "Tails". We need a **translator** to turn physical outcomes into numbers.

2. The Translator: Random Variable (X)

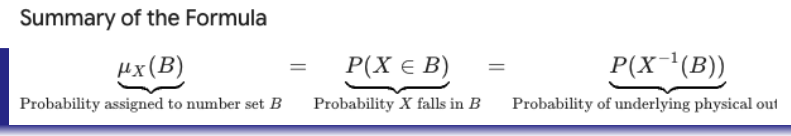
A random variable X translates real-world outcomes into numbers on the real line ($\mathbb{R}$):

- $X(\text{Heads}) = 1$
- $X(\text{Tails}) = 0$

3. The "Pushforward" Step-by-Step

Now suppose you ask: **"What is the probability that X equals 1?"**

1. **Target set on the number line (B):** You pick the set $B = \{1\}$.
2. **Look backward ($X^{-1}(B)$):** Which original outcome created the number 1?
   - Answer: **Heads.** (This set of original outcomes is the **preimage** $X^{-1}(B)$).
3. **Measure original probability (P):** What was the probability of getting Heads?
   - Answer: 0.5.
4. **Push it forward (μ_X):** You assign that 0.5 probability directly to the number 1 on the number line.


Summary of the Formula

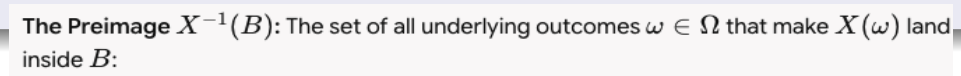
$$\underbrace{\mu_X(B)}_{\text{Probability assigned to number set } B} = \underbrace{P(X \in B)}_{\text{Probability } X \text{ falls in } B} = \underbrace{P(X^{-1}(B))}_{\text{Probability of underlying physical out}}$$

Random variable

A map X is **measurable** if for every Borel set $B \in \mathcal{B}(\mathbb{R})$, the preimage $X^{-1}(B) = \{\omega \in \Omega : X(\omega) \in B\} \in \mathcal{F}$.

A region of interest on the real number line (for example, the interval [2, 5]

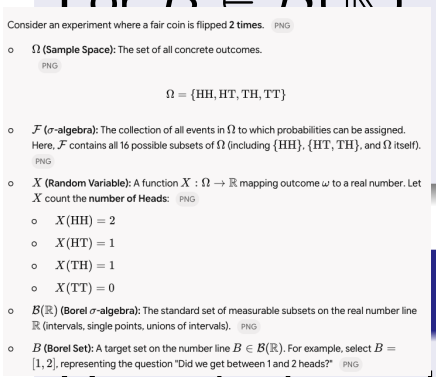
A **real-valued random variable** is a measurable map $X : (\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$.


The Preimage $X^{-1}(B)$: The set of all underlying outcomes $\omega \in \Omega$ that make $X(\omega)$ land inside B:

Law (distribution)

For $B \in \mathcal{B}(\mathbb{R})$, the law of X is the pushforward measure

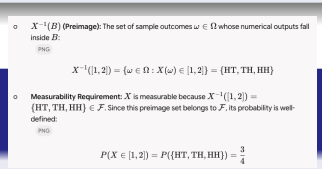
$$\mu_X(B) = \mathbb{P}(X \in B) = \mathbb{P}\big(X^{-1}(B)\big).$$


Consider an experiment where a fair coin is flipped **2 times.** PNG

- Ω **(Sample Space):** The set of all concrete outcomes. PNG

$$\Omega = \{\text{HH}, \text{HT}, \text{TH}, \text{TT}\}$$

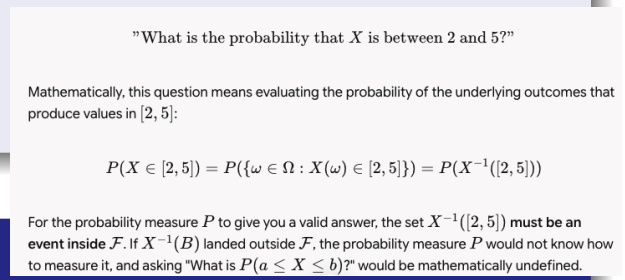
- $\mathcal{F}$ **(σ-algebra):** The collection of all events in Ω to which probabilities can be assigned. Here, $\mathcal{F}$ contains all 16 possible subsets of Ω (including $\{\text{HH}\}$, $\{\text{HT}, \text{TH}\}$, and Ω itself). PNG
- X **(Random Variable):** A function $X : \Omega \to \mathbb{R}$ mapping outcome ω to a real number. Let X count the **number of Heads:** PNG
  - $X(\text{HH}) = 2$
  - $X(\text{HT}) = 1$
  - $X(\text{TH}) = 1$
  - $X(\text{TT}) = 0$
- $\mathcal{B}(\mathbb{R})$ **(Borel σ-algebra):** The standard set of measurable subsets on the real number line $\mathbb{R}$ (intervals, single points, unions of intervals). PNG
- B **(Borel Set):** A target set on the number line $B \in \mathcal{B}(\mathbb{R})$. For example, select $B = [1, 2]$, representing the question "Did we get between 1 and 2 heads?" PNG


- $X^{-1}(B)$ **(Preimage):** The set of sample outcomes $\omega \in \Omega$ whose numerical outputs fall inside B: PNG

$$X^{-1}([1,2]) = \{\omega \in \Omega : X(\omega) \in [1,2]\} = \{\text{HT}, \text{TH}, \text{HH}\}$$

- **Measurability Requirement:** X is measurable because $X^{-1}([1,2]) = \{\text{HT}, \text{TH}, \text{HH}\} \in \mathcal{F}$. Since this preimage set belongs to $\mathcal{F}$, its probability is well-defined: PNG

$$P(X \in [1,2]) = P(\{\text{HT}, \text{TH}, \text{HH}\}) = \frac{3}{4}$$


"What is the probability that X is between 2 and 5?"

Mathematically, this question means evaluating the probability of the underlying outcomes that produce values in $[2, 5]$:

$$P(X \in [2,5]) = P(\{\omega \in \Omega : X(\omega) \in [2,5]\}) = P(X^{-1}([2,5]))$$

For the probability measure P to give you a valid answer, the set $X^{-1}([2,5])$ **must be an event inside $\mathcal{F}$.** If $X^{-1}(B)$ landed outside $\mathcal{F}$, the probability measure P would not know how to measure it, and asking "What is $P(a \le X \le b)$?" would be mathematically undefined.

Discrete (countable support; PMF p_X), absolutely continuous (PDF f_X; $F_X' = f_X$ a.e.), and mixed.

Why Random Variables are Useful?

Same information, but easier to use:

- A random variable $X : \Omega \to \mathbb{R}$ induces a distribution $\mathbb{P}_X$.
- Working with X or $\mathbb{P}_X$ encodes the same probabilistic information.
- $\Rightarrow$ Random variables let us write *equations* instead of handling sets and events.

2. General Events (Not Needing X)

An event does not have to be described using X. Any subset of Ω in $\mathcal{F}$ is an event:

- **"First die rolls a 4":**

$$A = \{(4,1),(4,2),(4,3),(4,4),(4,5),(4,6)\}$$

Advantages of random variables:

- *Algebraic manipulation:* can form $Y = 2X + 3$, $Z = X + Y$, etc.
- *Compounding:* joint r.v.s describe multiple processes easily.
- *Analytical tools:* enable expectations, variances, mgfs, characteristic functions.
- *Modeling:* regression, Bayesian models, stochastic processes all rely on r.v.s.

Key message: Random variables = distributions in disguise, but far more convenient for *calculation, composition, and modeling*.

Discrete random variables: PMF

What is a discrete RV?

X takes values in a countable set $A = \{x_1, x_2, \dots\}$. Probability mass sits on points; probabilities *add up* across values.

PMF (probability mass function)

The PMF $p_X(x) = \mathbb{P}(X = x)$ assigns a weight to each $x \in A$ with $p_X(x) \geq 0$ and $\sum_{x \in A} p_X(x) = 1$. For any measurable B,

$$\mathbb{P}(X \in B) = \sum_{x \in A \cap B} p_X(x).$$

Interpretation: p_X is a table (or bar chart) of *weights* at allowed values.

Examples: Bernoulli(p): $p(1) = p$, $p(0) = 1 - p$. Poisson(λ): $p(k) = e^{-\lambda}\, \lambda^k / k!$, $k = 0, 1, 2, \dots$.

Discrete random variables: CDF

CDF (cumulative distribution function)

$F_X(x) = \mathbb{P}(X \le x)$ accumulates mass up to x. For discrete X, F_X is a non-decreasing *step* function; at each support point x the jump size equals $p_X(x)$. Between support points F_X is flat.

Using the CDF

For $a < b$,

$$\mathbb{P}(a < X \le b) = F_X(b) - F_X(a), \qquad \mathbb{P}(X = x) = F_X(x) - F_X(x^-) = p_X(x).$$

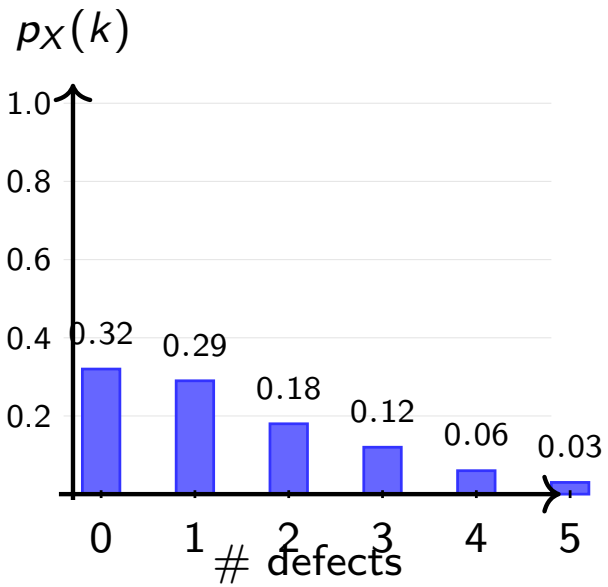
Properties: non-decreasing, right-continuous, $\lim_{x\to-\infty} F_X = 0$, $\lim_{x\to\infty} F_X = 1$.

When is the CDF meaningful?

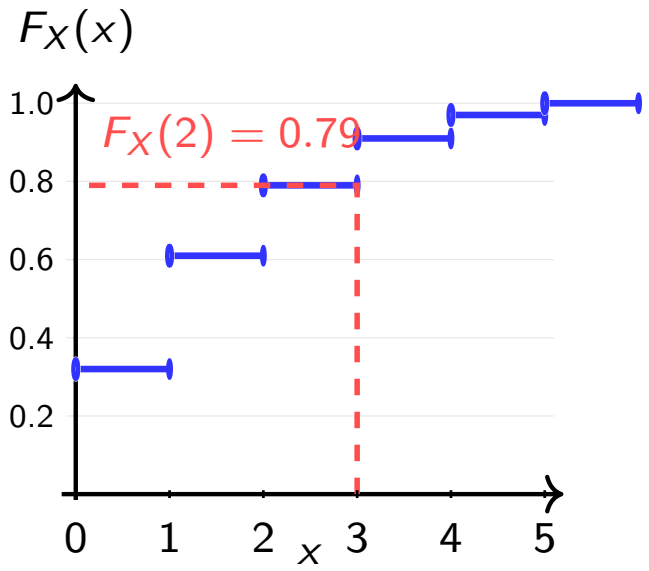
A CDF requires a meaningful order on the support. For nominal categories (unordered), prefer category probabilities (PMF); only define cumulative quantities once an inherent or agreed ordinal scale exists.

When the CDF *does* make sense (ordered counts)

PMF (Defect count)



CDF (cumulative steps)



Natural ordering

Values $0 < 1 < 2 < \ldots$ have clear meaning.
Events like $\{X \leq 2\}$ are interpretable.

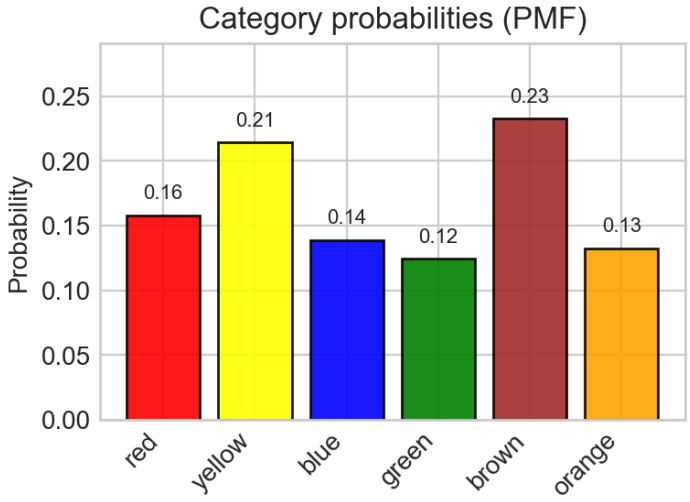
Meaningful cumulation

$F_X(k) = \mathbb{P}(X \leq k)$ answers concrete questions about defects.

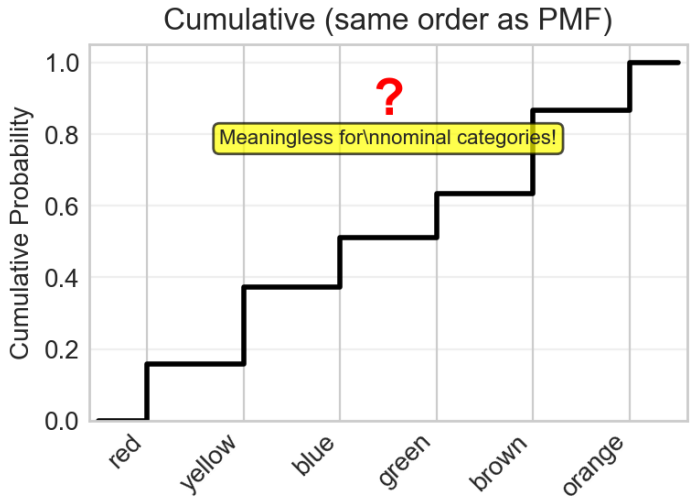
Key insight: Natural ordering $\Rightarrow$ meaningful cumulative probabilities

When CDF is *meaningless* (nominal categories)

PMF for M&M colors



Arbitrary "CDF" (alphabetical)



No natural order

Colors are *nominal categories*.
No meaningful $<$ relationship exists.

Why this fails

Different orderings give different "CDFs".
$\mathbb{P}(\text{color} \leq \text{Green})$ is nonsense!

Key insight: No natural order $\Rightarrow$ CDF is arbitrary and meaningless

Continuous random variables: PDF

What is a continuous RV?

No atoms: $\mathbb{P}(X = a) = 0$ for every $a \in \mathbb{R}$. Probabilities live on intervals/sets, not single points.

PDF (probability density function)

A nonnegative function f_X with $\int_{\mathbb{R}} f_X(x)\,dx = 1$ such that for any measurable B, $\mathbb{P}(X \in B) = \int_B f_X(x)\,dx$. For $a < b$,

$$\mathbb{P}(a < X \leq b) = \int_a^b f_X(x)\,dx.$$

Intuition: f_X is a *height*; probability is *area under the curve*. The PDF itself is not a probability.

Examples: Uniform(a, b): $f = 1/(b-a)$ on $[a, b]$; Exponential(λ): $\lambda e^{-\lambda x}$ for $x \geq 0$; Normal(μ, σ^2): bell-shaped.

Continuous random variables: CDF

CDF (cumulative distribution function)

$F_X(x) = \mathbb{P}(X \le x) = \int_{-\infty}^{x} f_X(t)\,dt$. For $a < b$,

$$\mathbb{P}(a < X \le b) = F_X(b) - F_X(a) = \int_a^b f_X(x)\,dx.$$

Properties: non-decreasing, right-continuous, $\lim_{x\to-\infty} F_X = 0$, $\lim_{x\to\infty} F_X = 1$.

Remember

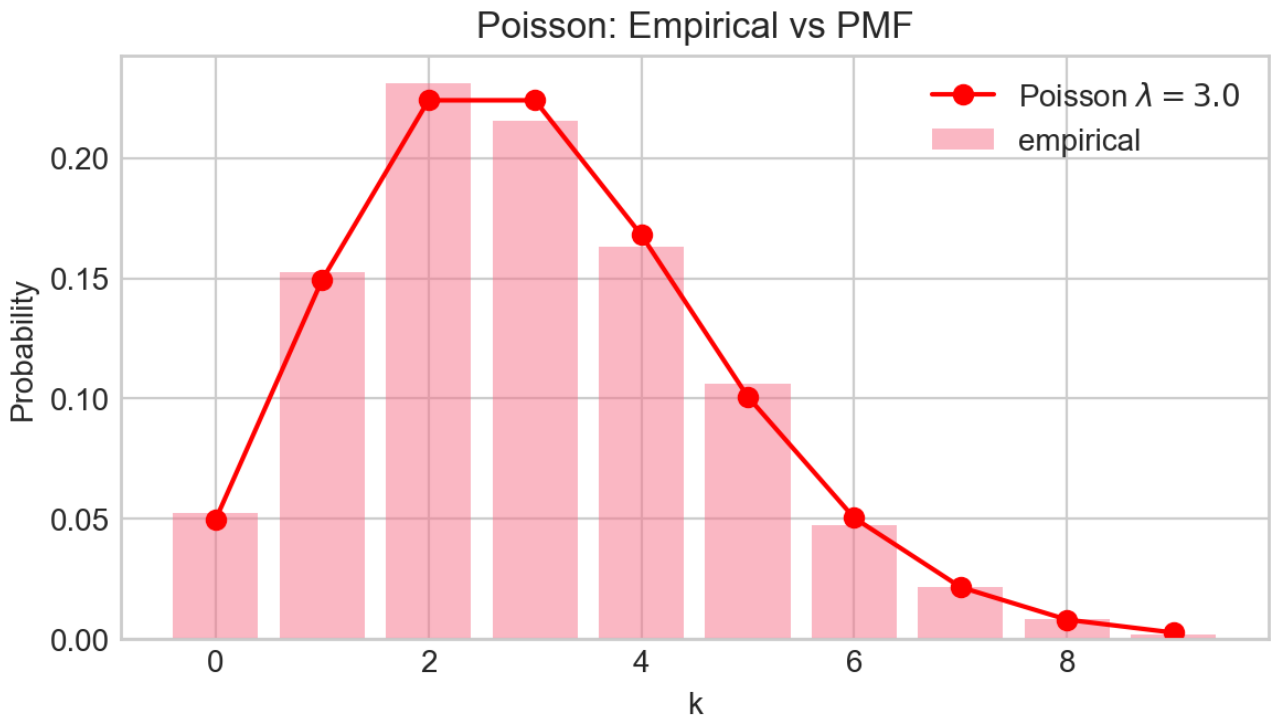
The PDF is the *derivative* of the CDF: $f_X(x) = F_X'(x)$ almost everywhere. The CDF is bounded in $[0, 1]$; the PDF is a *slope* and need not be ≤ 1 — probabilities come from *areas* under f_X, not heights.

Quantiles

For $q \in (0, 1)$, a q-quantile is any x with $F_X(x) \ge q$. If F_X is strictly increasing, the q-quantile is $F_X^{-1}(q)$.

Key Distributions (Discrete)

- Bernoulli(p): $\Pr(X = 1) = p$, $\Pr(X = 0) = 1 - p$, $\mathbb{E}[X] = p$, $\mathrm{Var}(X) = p(1 - p)$
- Binomial(n, p): sum of n iid Bernoulli; $\mathbb{E}[X] = np$, $\mathrm{Var}(X) = np(1 - p)$
- Poisson(λ): $\Pr(X = k) = e^{-\lambda}\lambda^k/k!$, $\mathbb{E}[X] = \lambda$, $\mathrm{Var}(X) = \lambda$

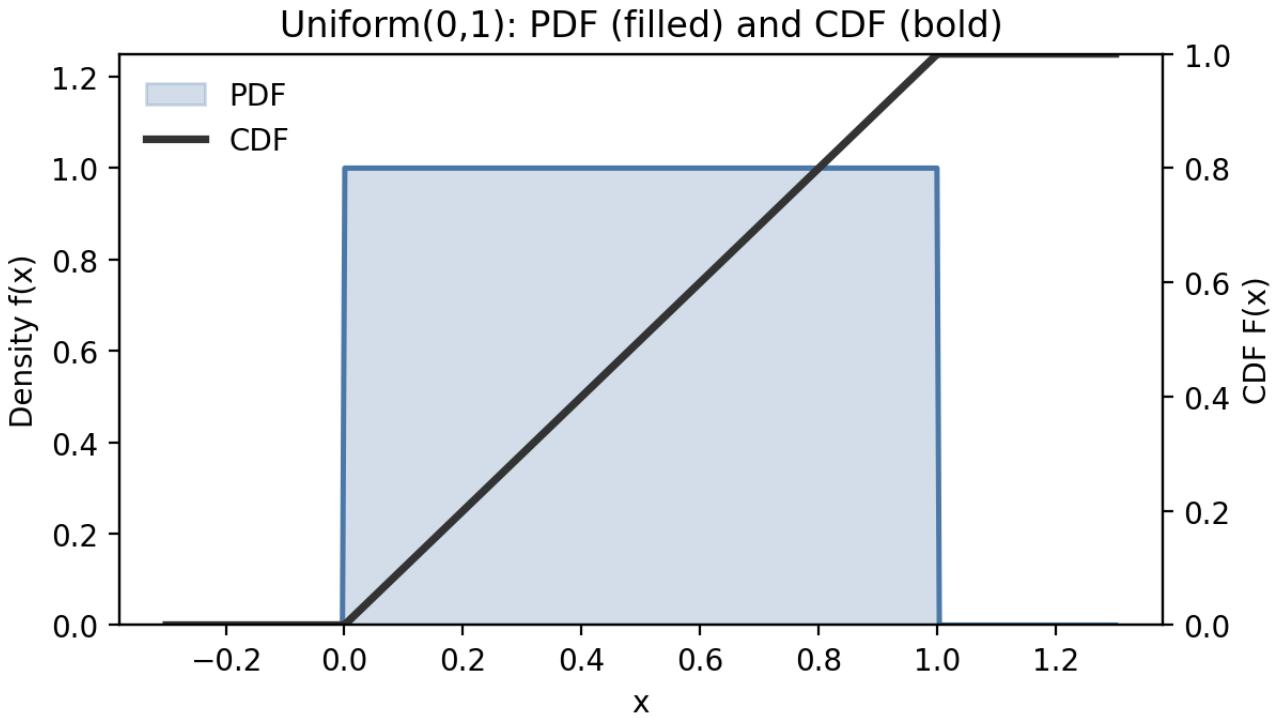


Uniform (a, b)

Formulas

$$\text{PDF: } f(x) = \begin{cases} \dfrac{1}{b-a}, & a \le x \le b, \\ 0, & \text{otherwise} \end{cases} \qquad \text{CDF: } F(x) = \begin{cases} 0, & x < a, \\ \dfrac{x-a}{b-a}, & a \le x \le b, \\ 1, & x > b \end{cases}$$

$$\text{Mean: } \mathbb{E}[X] = \frac{a+b}{2} \qquad \text{Variance: } \operatorname{Var}(X) = \frac{(b-a)^2}{12}$$



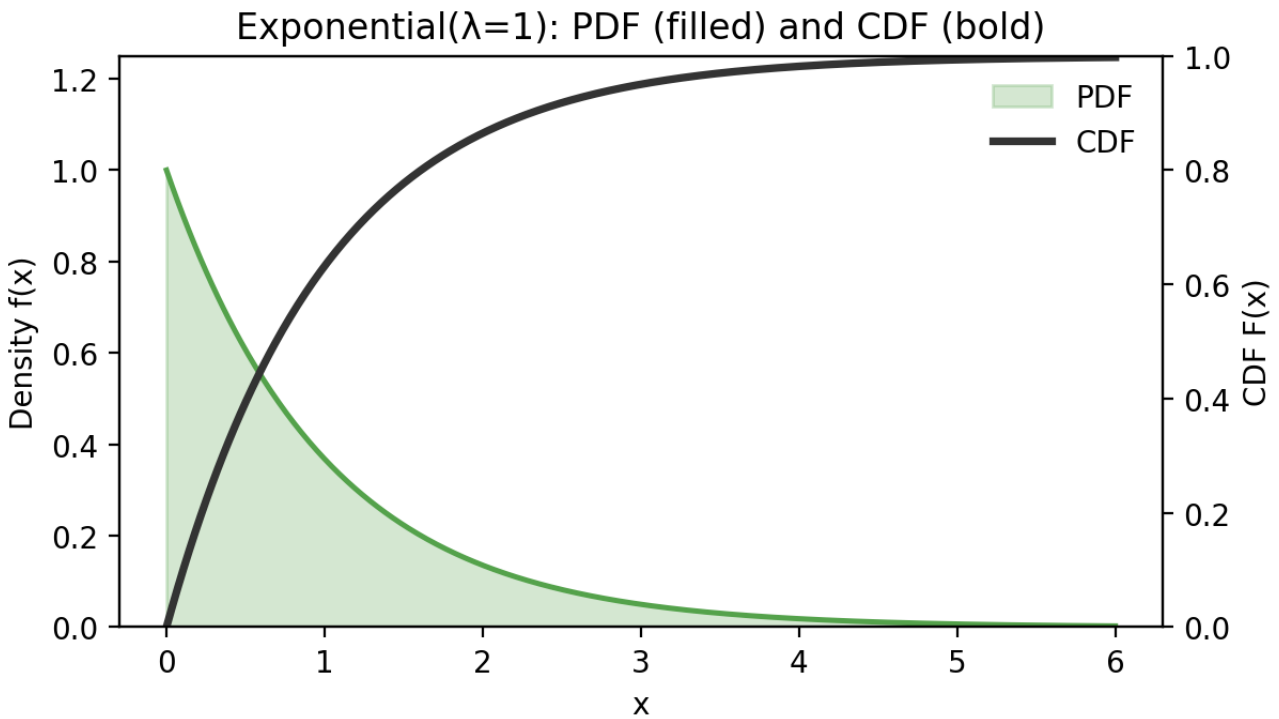
Exponential (λ)

Formulas

Support: $x \geq 0$ PDF: $f(x) = \lambda e^{-\lambda x}$ CDF: $F(x) = 1 - e^{-\lambda x}$

Memoryless: $\mathbb{P}(X > s + t \mid X > s) = e^{-\lambda t}$

Mean: $\mathbb{E}[X] = \dfrac{1}{\lambda}$ Variance: $\mathrm{Var}(X) = \dfrac{1}{\lambda^2}$

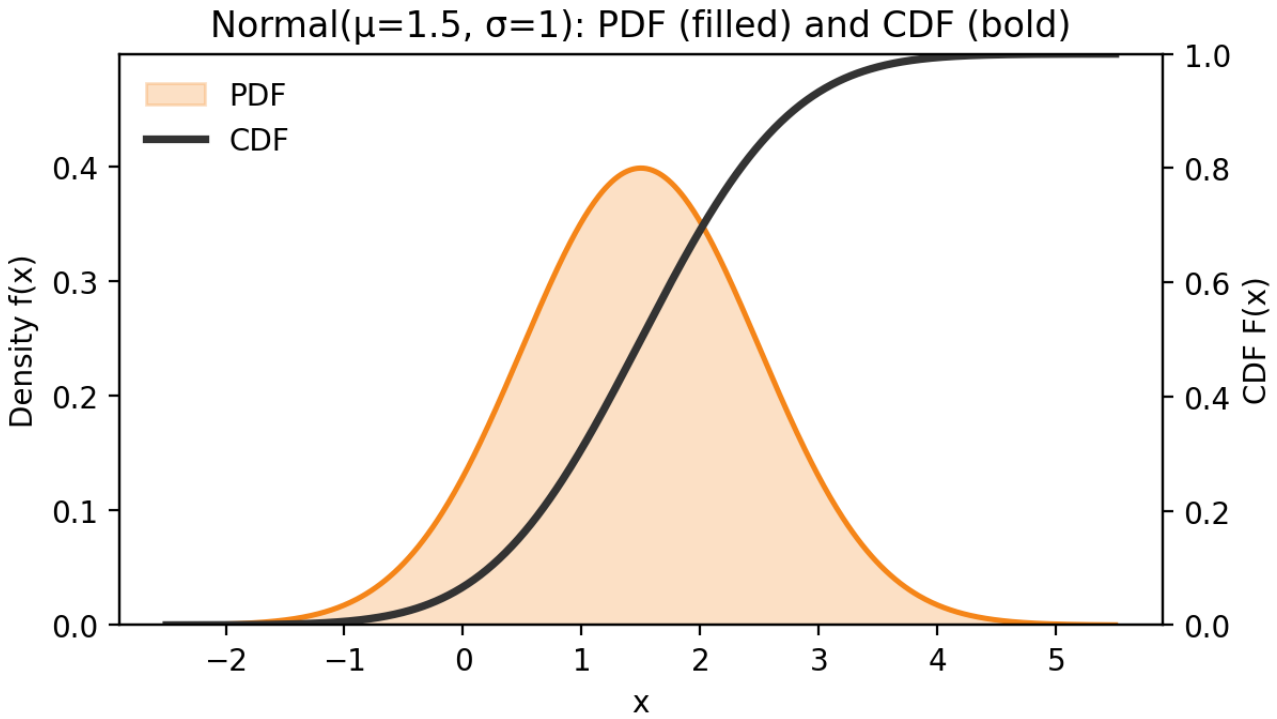


Normal (μ, σ^2)

Formulas

PDF: $f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2} \right)$ CDF: $F(x) = \Phi\left(\frac{x-\mu}{\sigma}\right)$ (no closed form)

Mean: $\mathbb{E}[X] = \mu$ Variance: $\mathrm{Var}(X) = \sigma^2$ Standardization: $\frac{X-\mu}{\sigma} \sim \mathcal{N}(0,1)$



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What is a Statistic?

Definition of a Statistic

- Let $X_1, \ldots, X_n$ be a sample of random variables taking values in $\mathcal{X}$.
- A **statistic** is a measurable function

$$T : \mathcal{X}^n \longrightarrow \mathcal{Y},$$

  where typically $\mathcal{Y} = \mathbb{R}^k$.
- For a realization $(x_1, \ldots, x_n)$, the statistic produces $T(x_1, \ldots, x_n)$.

Intuition:

- A statistic is a *summary* of the sample.
- It compresses n observations into a simpler object (number or vector).

Examples:

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i, \quad S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2, \quad \max(X_1, \ldots, X_n).$$

Descriptive vs Inferential Statistics

Descriptive statistics

- *Purpose:* summarize and present the features of the observed dataset; no claims beyond the data.
- *Mathematical view:* summary functionals $S(\mathbf{x})$ of a sample $\mathbf{x} = (x_1, \ldots, x_n)$ capturing location, spread, shape, and dependence.

Examples

- Mean $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$, median, mode.
- Variance $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$, standard deviation, range, IQR.
- Counts/proportions (categorical), histograms, boxplots, scatter plots.

Inferential statistics

- *Purpose:* generalize from the sample to the population using probability models; quantify uncertainty.
- *Outputs:* estimates with standard errors, confidence intervals, hypothesis tests, predictive statements.

Examples

- Confidence intervals; hypothesis tests (e.g., t-test); p-values.
- Parametric/GLM/Regression inference; model comparison/selection.
- Bootstrap-based uncertainty quantification.

Key distinction: *descriptive* = what this dataset looks like, *inferential* = what we can say about the population (with uncertainty).

Expectation via measurable functions

Discrete case (PMF p_X)

For measurable g with $\mathbb{E}[|g(X)|] < \infty$,

Lebesgue measure (dx): The standard measure of "length" on the real line $\mathbb{R}$.

$$\mathbb{E}[g(X)] = \sum_{x \in \text{supp}(X)} g(x)\, p_X(x) \quad \text{(absolute conv}$$

Tưởng tượng trục số x là một con đường dài từ $-\infty$ đến $+\infty$:

- dx **là độ dài bước chân:** Khi bạn viết $\int \ldots dx$, nghĩa là bạn cho một thước đo bước từng bước nhỏ có độ dài dx chạy từ đầu đường đến cuối đường.
- $d\mu(x)$ **là lượng xác suất rải trên bước chân đó:** Tại bước chân dx này, có bao nhiêu vàng/xác suất rơi ở đây?

Nếu bạn chỉ lấy giá trị $g(x)$ nhân với dx rồi cộng lại ($\int g(x)dx$), bạn đang **CỘNG ĐỀU** mọi khoảng dx trên con đường với vai trò ngang nhau. Điều này chỉ đúng nếu xác suất được rải đều tâm tập từ $-\infty$ đến $+\infty$ (điều này còn không phải là một phân bố xác suất hợp lệ vì tổng xác suất sẽ ra vô hạn).

In particular, take $g(x) = x$ to get $\mathbb{E}[X] = \sum_x x\, p_X(x)$.

Continuous case (PDF f_X)

Lebesgue measure (dx): The standard measure of "length" on the real line $\mathbb{R}$.

If X admits a density f_X w.r.t. Lebesgue measure and $\mathbb{E}[|g(X)|] < \infty$,

if we dont use this, we have to transform the variable

LOTUS

$$\mathbb{E}[g(X)] = \int_{\mathbb{R}} g(x)\, f_X(x)\, dx.$$

In general, with law μ_X, $\mathbb{E}[g(X)] = \int g\, d\mu_X$. Setting $g(x) = x$ yields $\mathbb{E}[X]$.

Note: $\mathbb{E}[g(X)]$ is defined whenever $\int |g|\, d\mu_X < \infty$.

Expectation and variance: definitions and

Thông thường, để tính trung bình của một lượng mới $Y = g(X)$, lý thuyết bắt bạn phải:

1. Tìm hàm mật độ xác suất mới $f_Y(y)$ của Y.
2. Tính $\int y f_Y(y) dy$.

Tuy nhiên, công thức trên (được gọi là *Luật của nhà thống kê vô tư* - LOTUS) cho phép bạn **bỏ qua bước tìm** $f_Y(y)$. Bạn có thể tính trực tiếp bằng cách giữ nguyên hàm mật độ gốc $f_X(x)$ của X và chỉ việc biến đổi giá trị đầu vào thành $g(x)$. PNG

Definitions

For integrable X, the mean of the random variable is defined as

$$\mathbb{E}[X] = \int_\Omega X \, d\mathbb{P} = \int_{\mathbb{R}} x \, d\mu_X(x)$$

If $\mathbb{E}[X^2] < \infty$, then the variance of the random variable is defined as

$$\mathrm{Var}(X) = \mathbb{E}\big[(X - \mathbb{E}[X])^2\big]$$

Algebraic properties

- Linearity: $\mathbb{E}[aX + b] = a\,\mathbb{E}[X] + b$ and $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$.
- Affine scaling: $\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X)$; non-negativity: $\mathrm{Var}(X) \geq 0$ with equality iff X is a.s. constant.
- Second-moment identity: $\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$ when $\mathbb{E}[X^2] < \infty$.

Higher-Order Moments: Skewness and Kurtosis

Definitions (when finite)

Let $\mu = \mathbb{E}[X]$, $\sigma^2 = \mathrm{Var}(X)$, and central moments $\mu_k = \mathbb{E}\big[(X-\mu)^k\big]$.
Skewness $\gamma_1 := \mu_3/\sigma^3$; Kurtosis $\beta_2 := \mu_4/\sigma^4$; Excess kurtosis $\gamma_2 := \beta_2 - 3$.

Relevance and interpretation

- Skewness (γ_1): *asymmetry*. $\gamma_1 > 0$ right-skewed; $\gamma_1 < 0$ left-skewed; $\gamma_1 = 0$ for symmetric laws.
- Kurtosis (β_2) and excess (γ_2): tail weight/peakedness. Normal: $\beta_2 = 3$ hence $\gamma_2 = 0$.
- Practice: diagnose non-normality, tail risk, outlier propensity; both are outlier-sensitive (kurtosis especially).
- Sampling: for sample means, standardized skew decays $\propto n^{-1/2}$ and excess kurtosis $\propto n^{-1}$ (i.i.d., finite moments).

Example: Bernoulli (p)

Random variable and summary

Let $X \sim \text{Bernoulli}(p)$ with support $\{0, 1\}$, $\mathbb{P}(X = 1) = p$, $\mathbb{P}(X = 0) = 1 - p$. Then $\mathbb{E}[X] = p$ and $\text{Var}(X) = p(1 - p)$.

Computation.

$$\mathbb{E}[X] = \sum_{x \in \{0,1\}} x \, p_X(x) = 0 \cdot (1 - p) + 1 \cdot p = p.$$

$$\mathbb{E}[X^2] = \sum_{x \in \{0,1\}} x^2 \, p_X(x) = 0^2 \cdot (1 - p) + 1^2 \cdot p = p \quad (\text{since } X^2 = X).$$

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = p - p^2 = p(1 - p).$$

Example: Exponential (λ)

Random variable and summary

Let $X \sim \text{Exponential}(\lambda)$ with density $f_X(x) = \lambda e^{-\lambda x}\,\mathbf{1}_{\{x \geq 0\}}$. Then $\mathbb{E}[X] = \dfrac{1}{\lambda}$ and $\text{Var}(X) = \dfrac{1}{\lambda^2}$.

Computation.

$$\mathbb{E}[X] = \int_0^\infty x\, f_X(x)\, dx = \int_0^\infty x\, \lambda e^{-\lambda x}\, dx = \frac{1}{\lambda} \quad \text{(IBP or } \Gamma \text{ function)}.$$

$$\mathbb{E}[X^2] = \int_0^\infty x^2\, f_X(x)\, dx = \int_0^\infty x^2\, \lambda e^{-\lambda x}\, dx = \frac{2}{\lambda^2}.$$

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}.$$

Example: Normal (μ, σ^2)

Random variable and summary

Let $X \sim \mathcal{N}(\mu, \sigma^2)$ with density $f_X(x) = \dfrac{1}{\sigma\sqrt{2\pi}} \exp\Big(- \dfrac{(x-\mu)^2}{2\sigma^2} \Big)$. Then $\mathbb{E}[X] = \mu$ and $\mathrm{Var}(X) = \sigma^2$.

Computation. Let $Z = \dfrac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1)$. Then

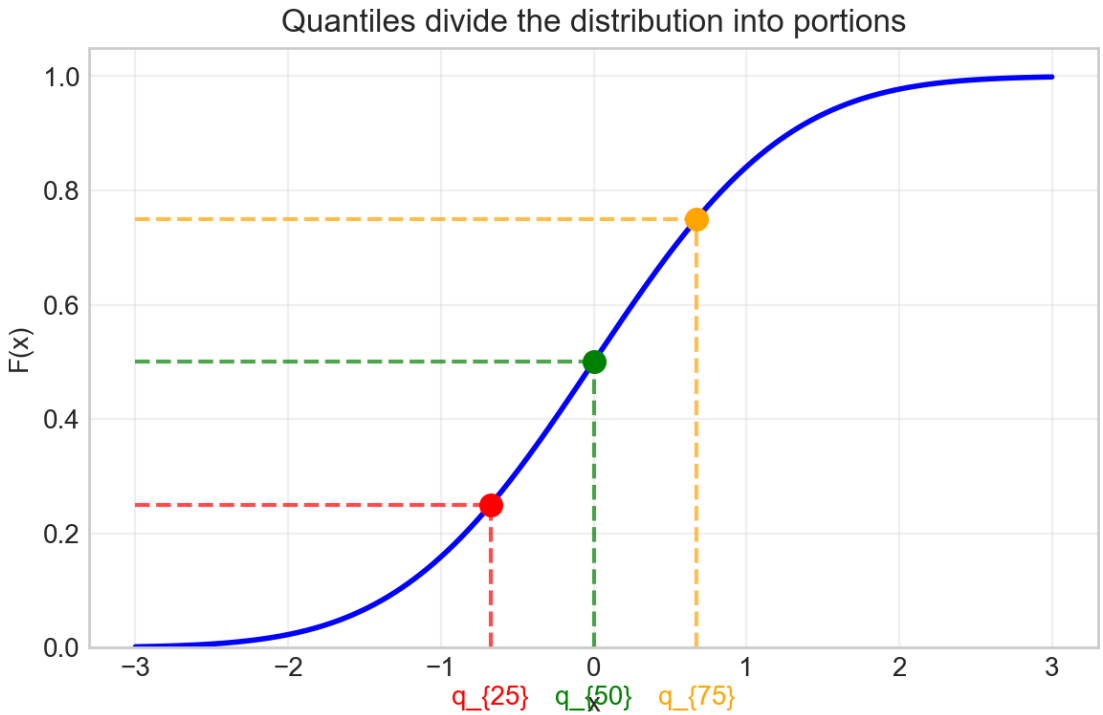
$$\mathbb{E}[X] = \mathbb{E}[\mu + \sigma Z] = \mu + \sigma\, \mathbb{E}[Z] = \mu,$$
$$\mathrm{Var}(X) = \mathrm{Var}(\mu + \sigma Z) = \sigma^2\, \mathrm{Var}(Z) = \sigma^2 \quad (\mathrm{Var}(Z) = 1).$$

Definition of Quantiles

Quantile

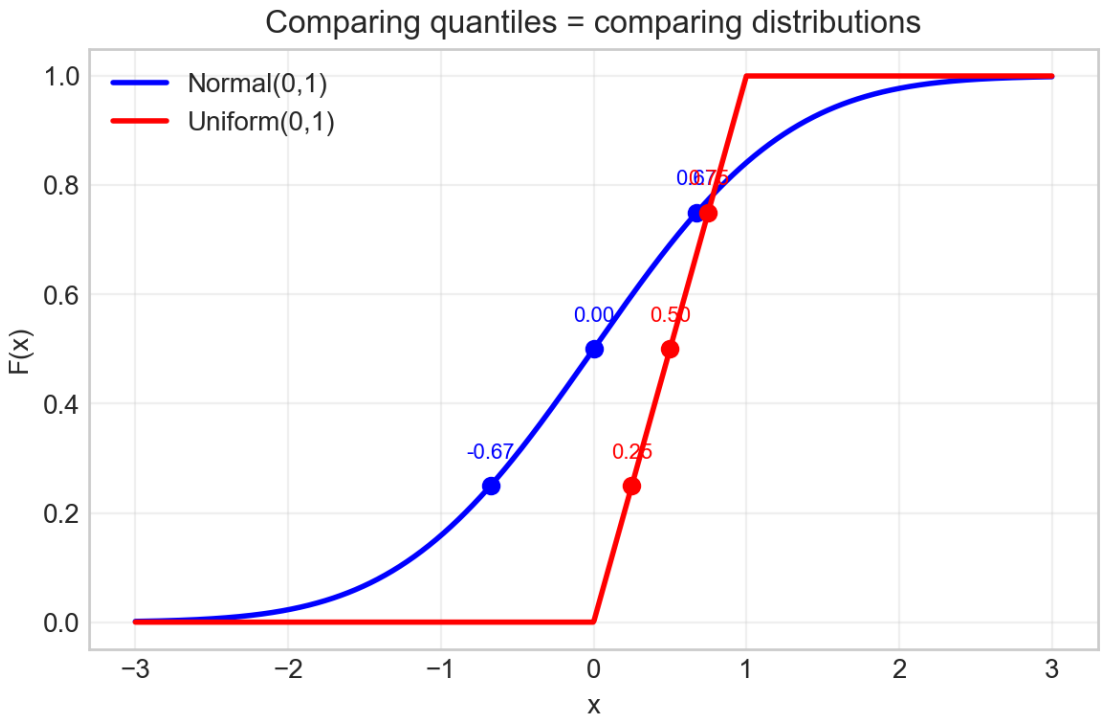
The p-quantile is $q_p = \inf\{x : F(x) \geq p\}$, where F is the CDF.

Intuition: "the value below which a fraction p of the data lies."



Why are quantiles useful?

- Quantiles summarize distributional shape (center, spread, tails).
- Robust to outliers (median, quartiles).
- Natural way to compare distributions.



Definition of QQ-Plot

QQ-Plot

QQ-plot = plot quantiles of one distribution against quantiles of another.

- If distributions are the same $\rightarrow$ points lie on line $y = x$.
- Deviations show differences in location, scale, or tails.

Bước 1: Sắp xếp dữ liệu thực nghiệm (Order Statistics)

Cho tập mẫu $X = \{x_1, x_2, \ldots, x_n\}$ gồm n quan sát. Sắp xếp các giá trị theo thứ tự tăng dần:

$$x_{(1)} \leq x_{(2)} \leq \cdots \leq x_{(n)}$$

Bước 2: Gán vị trí xác suất tích lũy (p_i)

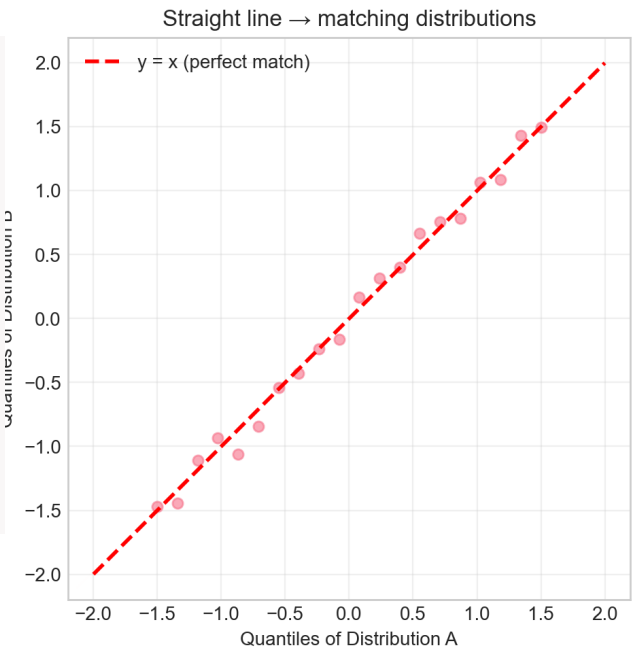
Vạch các mốc xác suất $p_i \in (0, 1)$ đại diện cho tỷ lệ phần trăm dữ liệu nằm phía bên trái điểm thứ i:

$$p_i = \frac{i - 0.5}{n} \quad (\text{với } i = 1, 2, \ldots, n)$$

Bước 3: Tính phân vị lý thuyết (q_i)

Dùng hàm nghịch đảo CDF (Percent Point Function - PPF) của phân bố mục tiêu (ví dụ phân bố Chuẩn chuẩn hóa $\mathcal{N}(0, 1)$) để quy đổi p_i thành tọa độ lý thuyết:

$$q_i = \Phi^{-1}(p_i)$$



Bước 4: Dựng đường tham chiếu bền vững (Robust Fit Line pass through Q_1, Q_3)

Để tránh nhiễu do điểm ngoại lệ ở hai đuôi, đường thẳng tham chiếu $y = m \cdot q + c$ được xác định dựa trên hai điểm tứ phân vị 25% (Q_1) và 75% (Q_3):

1. **Tọa độ lý thuyết:** $x_{Q1} = \Phi^{-1}(0.25) \approx -0.6745$ và $x_{Q3} = \Phi^{-1}(0.75) \approx +0.6745$.
2. **Tọa độ thực nghiệm:** $y_{Q1} = \text{Percentile}(X, 25)$ và $y_{Q3} = \text{Percentile}(X, 75)$.
3. **Hệ số góc (Slope):**

$$m = \frac{y_{Q3} - y_{Q1}}{x_{Q3} - x_{Q1}} = \frac{y_{Q3} - y_{Q1}}{1.349}$$

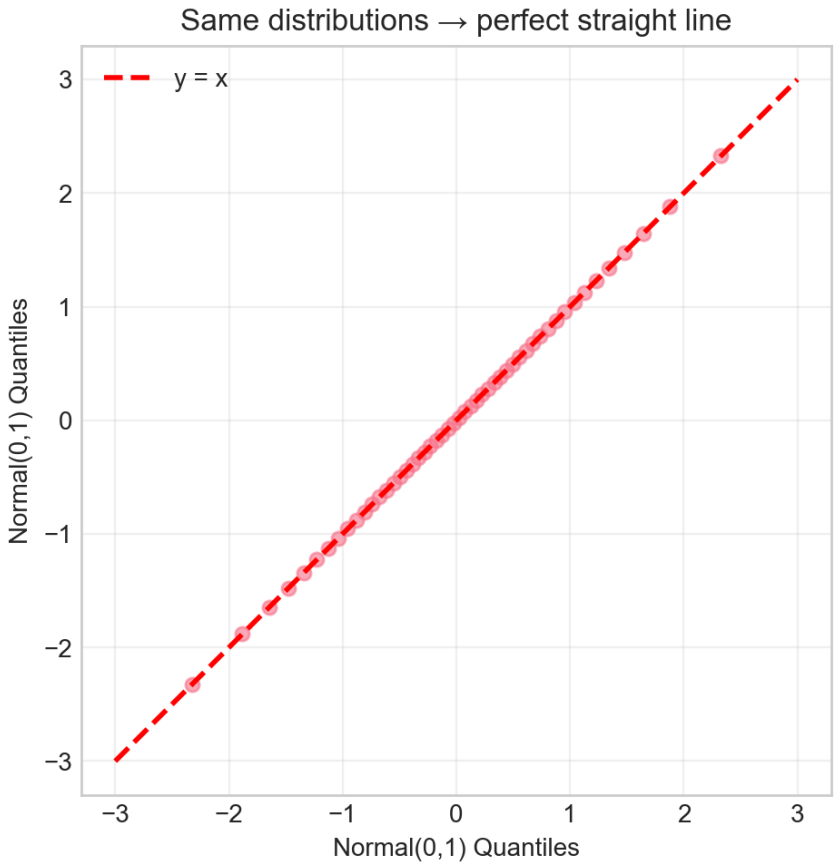
4. **Tung độ gốc (Intercept):**

$$c = y_{Q1} - m \cdot x_{Q1}$$

Bước 5: Chấm tọa độ và chuẩn đoán

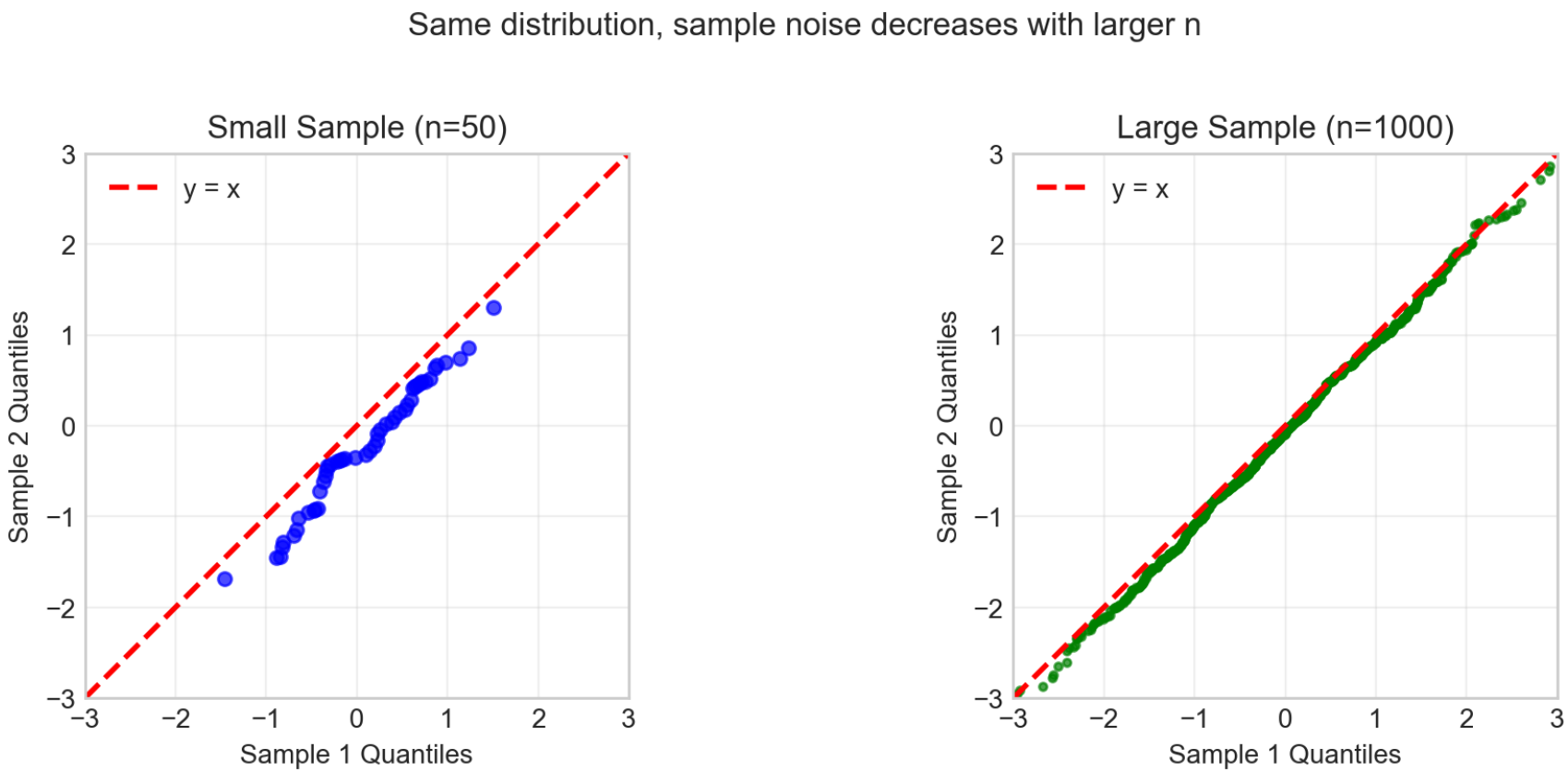
- Biểu diễn các cặp điểm $(q_i, x_{(i)})$ lên mặt phẳng tọa độ.
- Vẽ đường thẳng $y = m \cdot q + c$.
- **Đánh giá:** Nếu các điểm $(q_i, x_{(i)})$ bám sát đường thẳng, dữ liệu tuân theo phân bố mục tiêu. Các độ lệch dạng chữ S, uốn cong Parabol thể hiện đặc tính đuôi dầy, đuôi mỏng hoặc độ lệch (skewness).

QQ-Plot: Normal vs Normal (Theoretical)



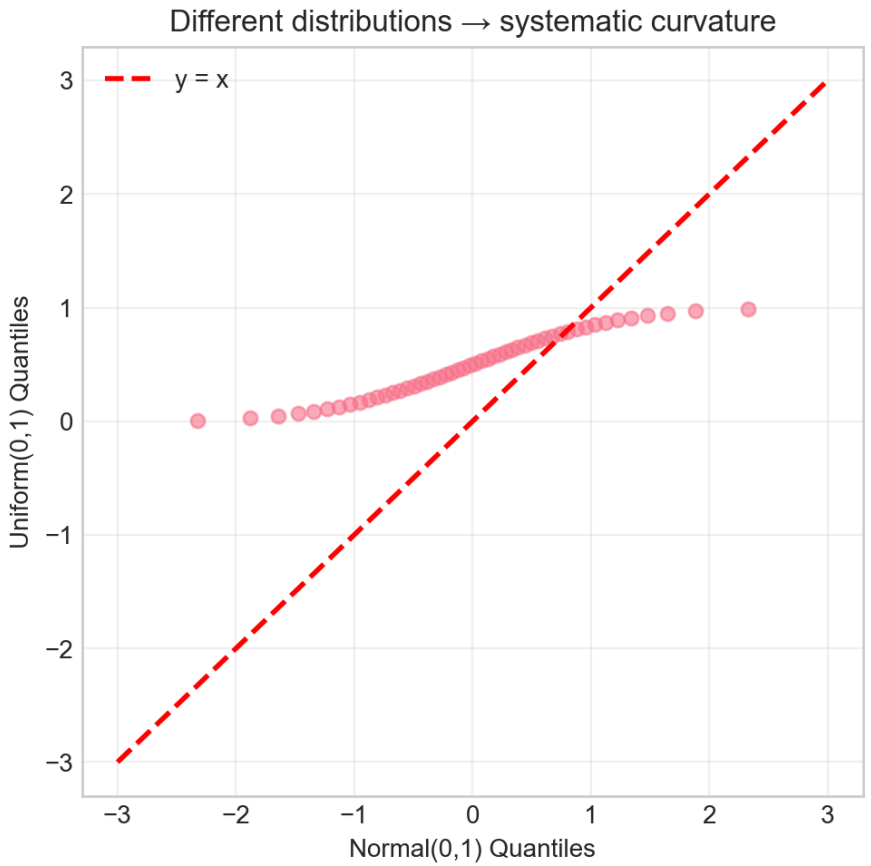
Same distributions → perfect straight line.

QQ-Plot: Normal vs Normal (Empirical)



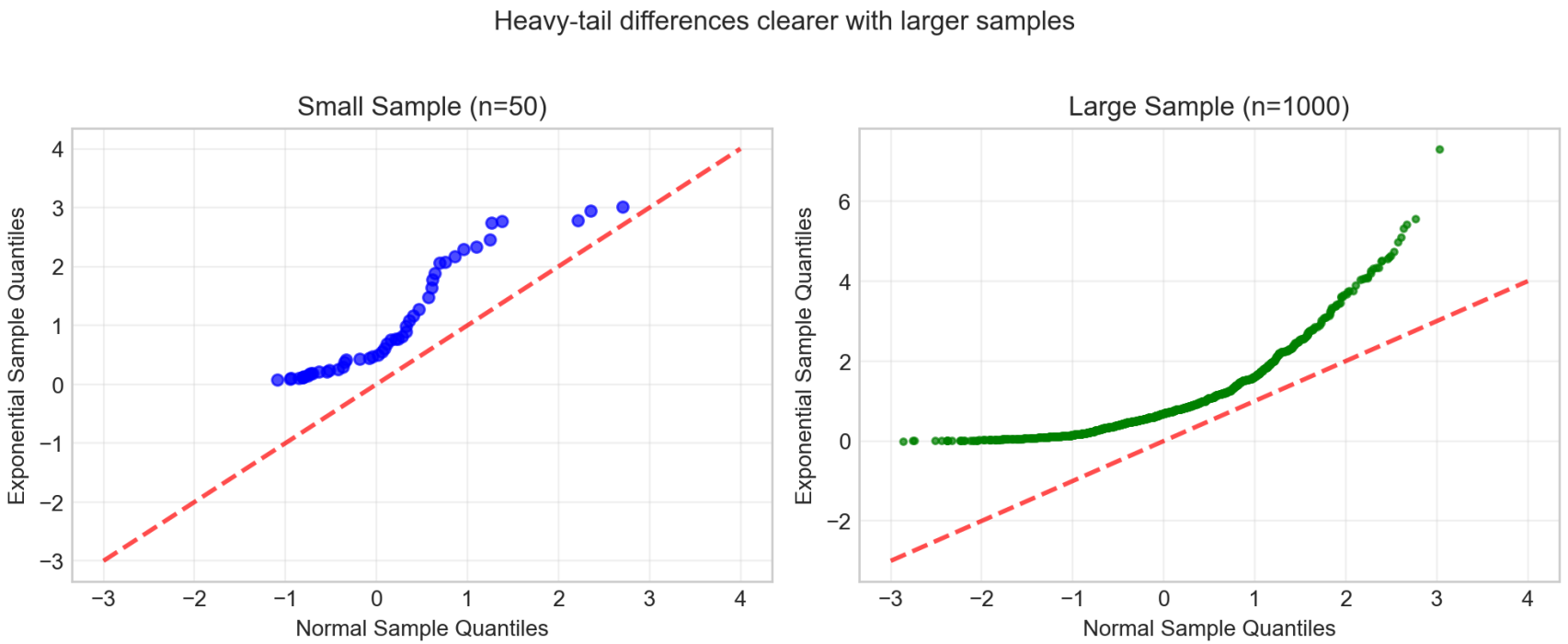
Same distribution: larger samples reduce noise, approach theoretical line.

QQ-Plot: Normal vs Uniform (Theoretical)



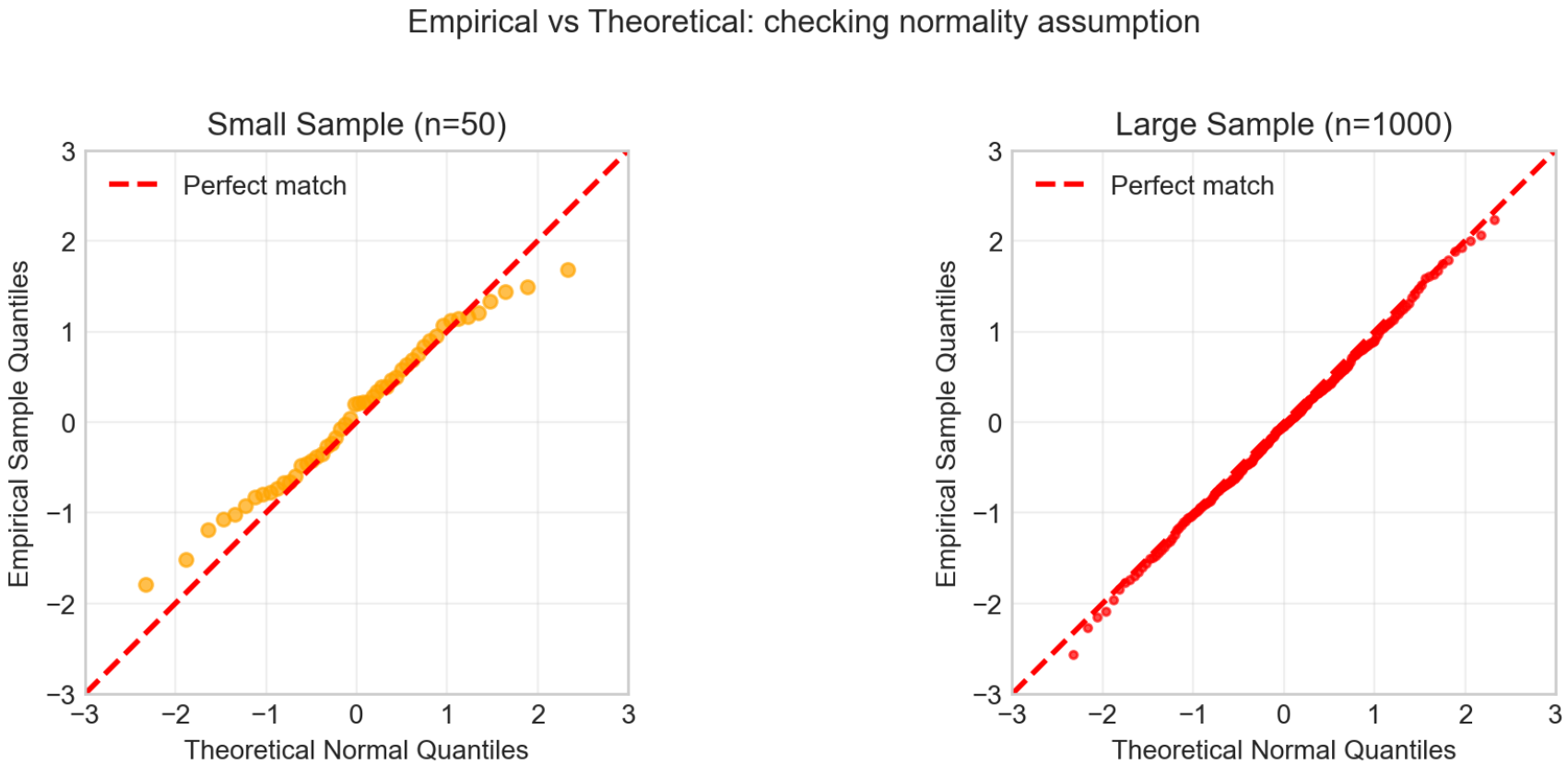
Different distributions → systematic curvature.

QQ-Plot: Normal vs Exponential (Empirical)



Different distributions: larger samples reveal clearer patterns.

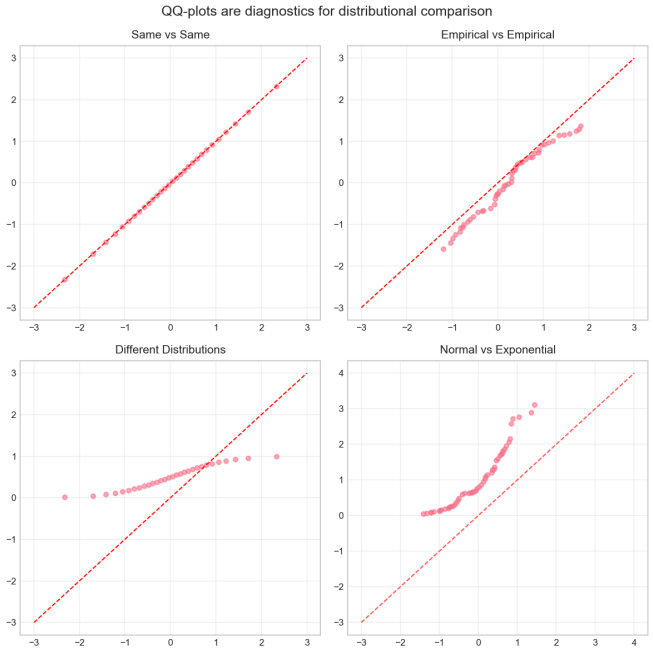
QQ-Plot: Empirical vs Theoretical Normal



Testing normality assumption: straight line = good fit to normal distribution.

Why QQ-Plots Matter

- Tool to check if a dataset matches a theoretical distribution (empirical vs theoretical).
- Tool to compare two datasets (empirical vs empirical).
- Straight line $\rightarrow$ matching distributions.
- Curvature or deviation $\rightarrow$ differences in location, spread, or tails.



Moment Generating Function (MGF)

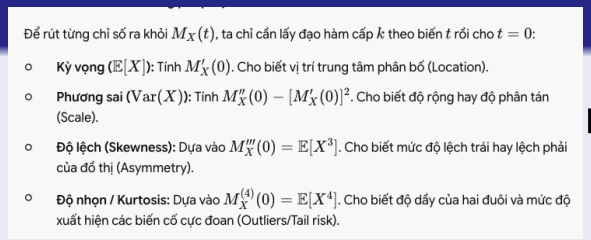
Definition

For a real-valued random variable X, the moment generating function (when finite) is

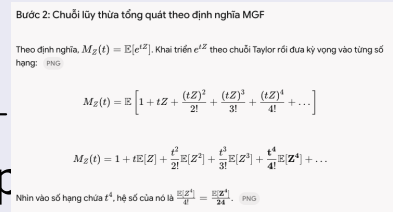
$$M_X(t) = \mathbb{E}\left[e^{tX}\right], \qquad t \in D := \{t \in \mathbb{R} : \mathbb{E}[e^{tX}] < \infty\}.$$

Key properties and relevance

- Normalization: $M_X(0) = 1$. If M_X exists on D with $0 \in \text{int}(D)$, it uniquely determines the law of X.
- Moments: whenever $\mathbb{E}[|X|^k] < \infty$, $M_X^{(k)}(0) = \mathbb{E}[X^k]$.
- Affine and sums: $M_{aX+b}(t) = e^{bt}\, M_X(at)$; if $X \perp\!\!\!\perp Y$, then $M_{X+Y}(t) = M_X(t)\, M_Y(t)$.
- Uses: identify distributions (e.g., Normal $M_X(t) = \exp(\mu t +$ compute moments, and obtain distributions of sums of indep variables.
- Caveat: may fail to exist ne y-tailed laws (e.g., lognormal has $M_X(t) = \infty$ for $t > 0$).


Để rút từng chỉ số ra khỏi $M_X(t)$, ta chỉ cần lấy đạo hàm cấp k theo biến t rồi cho $t = 0$:

- **Kỳ vọng ($\mathbb{E}[X]$):** Tính $M'_X(0)$. Cho biết vị trí trung tâm phân bố (Location).
- **Phương sai ($\text{Var}(X)$):** Tính $M''_X(0) - [M'_X(0)]^2$. Cho biết độ rộng hay độ phân tán (Scale).
- **Độ lệch (Skewness):** Dựa vào $M'''_X(0) = \mathbb{E}[X^3]$. Cho biết mức độ lệch trái hay lệch phải của đồ thị (Asymmetry).
- **Độ nhọn / Kurtosis:** Dựa vào $M^{(4)}_X(0) = \mathbb{E}[X^4]$. Cho biết độ dầy của hai đuôi và mức độ xuất hiện các biến cố cực đoan (Outliers/Tail risk).

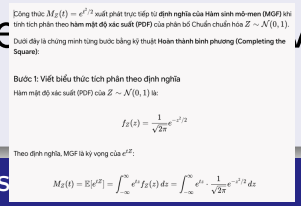

Bước 2: Chuỗi lũy thừa tổng quát theo định nghĩa MGF

Theo định nghĩa, $M_Z(t) = \mathbb{E}[e^{tZ}]$. Khai triển e^{tZ} theo chuỗi Taylor rồi đưa kỳ vọng vào từng số hạng: PNG

$$M_Z(t) = \mathbb{E}\left[1 + tZ + \frac{(tZ)^2}{2!} + \frac{(tZ)^3}{3!} + \frac{(tZ)^4}{4!} + \dots\right]$$

$$M_Z(t) = 1 + t\mathbb{E}[Z] + \frac{t^2}{2!}\mathbb{E}[Z^2] + \frac{t^3}{3!}\mathbb{E}[Z^3] + \frac{t^4}{4!}\mathbb{E}[Z^4] + \dots$$

Nhìn vào số hạng chứa t^4, hệ số của nó là $\frac{\mathbb{E}[Z^4]}{4!} = \frac{\mathbb{E}[Z^4]}{24}$. PNG


Công thức $M_Z(t) = e^{t^2/2}$ xuất phát trực tiếp từ định nghĩa của Hàm sinh mô-men (MGF) khi tính tích phân theo hàm mật độ xác suất (PDF) của phân bố Chuẩn hóa $Z \sim \mathcal{N}(0,1)$.

Dưới đây là chứng minh từng bước bằng kỹ thuật Hoàn thành bình phương (Completing the Square):

Bước 1: Viết biểu thức tích phân theo định nghĩa

Hàm mật độ xác suất (PDF) của $Z \sim \mathcal{N}(0,1)$ là:

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

Theo định nghĩa, MGF là kỳ vọng của e^{tZ}:

$$M_Z(t) = \mathbb{E}[e^{tZ}] = \int_{-\infty}^{\infty} e^{tz} f_Z(z)\, dz = \int_{-\infty}^{\infty} e^{tz} \cdot \frac{1}{\sqrt{2\pi}} e^{-z^2/2}\, dz$$

Characteristic Function

Definition

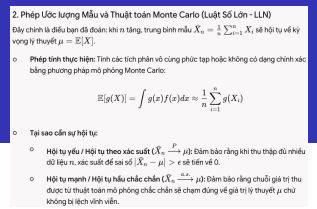
The characteristic function of X is

$$\varphi_X(t) = \mathbb{E}\left[e^{itX}\right], \qquad t \in \mathbb{R}.$$

Key properties and relevan

- Always exists; $\varphi_X(0) = 1$, $|\varphi_X(t)| \le 1$; uniformly continuous and positive-definite.
- Uniqueness and inversion: φ_X uniquely determines the law; inversion formulas recover the CDF/PDF under mild regularity.
- Moments: if $\mathbb{E}[|X|^k] < \infty$, then $\varphi_X^{(k)}(0) = i^k\,\mathbb{E}[X^k]$.
- Affine and sums: $\varphi_{aX+b}(t) = e^{ibt}\,\varphi_X(at)$; if $X \perp\!\!\!\perp Y$, then $\varphi_{X+Y}(t) = \varphi_X(t)\,\varphi_Y(t)$.
- Limit theory: Lévy continuity theorem links pointwise convergence of φ_{X_n} to $X_n \xrightarrow{\mathcal{D}} X$; core tool for CLT and stable laws.

Why Convergence Matters in Probability


2. Phép Ước lượng Mẫu và Thuật toán Monte Carlo (Luật Số Lớn - LLN)

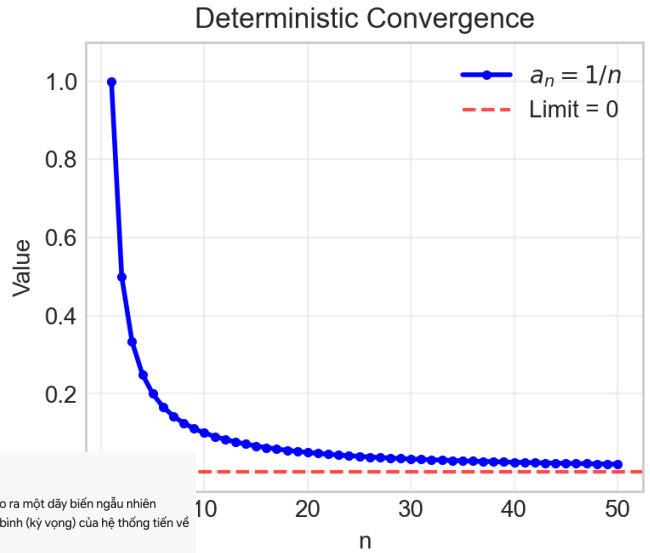
Đây chính là điều bạn đã đoán: khi n tăng, trung bình mẫu $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$ sẽ hội tụ về kỳ vọng lý thuyết $\mu = \mathbb{E}[X]$.

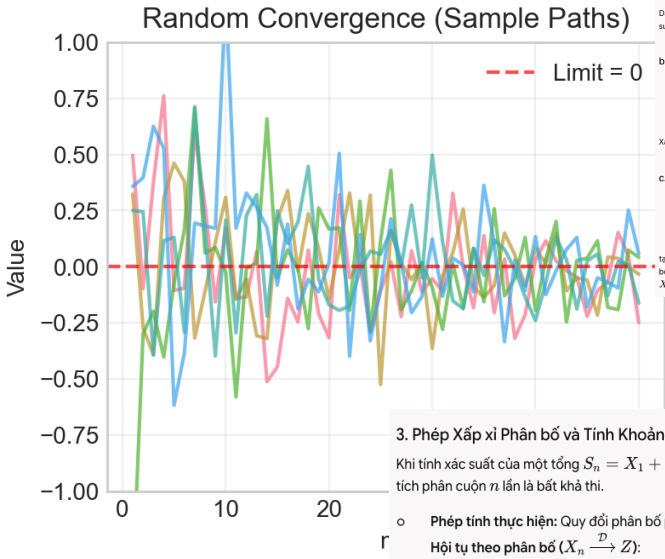
- **Phép tính thực hiện:** Tính các tích phân vô cùng phức tạp hoặc không có dạng chính xác bằng phương pháp mô phỏng Monte Carlo:

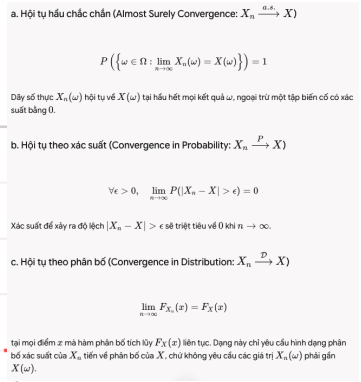
$$\mathbb{E}[g(X)] = \int g(x)f(x)dx \approx \frac{1}{n}\sum_{i=1}^{n} g(X_i)$$

- **Tại sao cần sự hội tụ:**
  - **Hội tụ yếu / Hội tụ theo xác suất** ($\bar{X}_n \xrightarrow{P} \mu$): Đảm bảo rằng khi thu thập đủ nhiều dữ liệu n, xác suất để sai số $|\bar{X}_n - \mu| > \epsilon$ sẽ tiến về 0.
  - **Hội tụ mạnh / Hội tụ hầu chắc chắn** ($\bar{X}_n \xrightarrow{a.s.} \mu$): Đảm bảo rằng chuỗi giá trị thu được từ thuật toán mô phỏng chắc chắn sẽ chạm đúng về giá trị lý thuyết μ chứ không bị lệch vĩnh viễn.

- Deterministic sequences: $a_n \to a$ means terms get arbitrarily close to a fixed limit.
- For random variables (X_n), multiple notions of *getting closer* exist.
- Hierarchy: a.s. $\Rightarrow$ in probability $\Rightarrow$ in distribution (no converses in general).






a. Hội tụ hầu chắc chắn (Almost Surely Convergence: $X_n \xrightarrow{a.s.} X$)

$$P\left(\left\{\omega \in \Omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\right\}\right) = 1$$

Dãy số thực $X_n(\omega)$ hội tụ về $X(\omega)$ tại hầu hết mọi kết quả ω, ngoại trừ một tập biến cố có xác suất bằng 0.

b. Hội tụ theo xác suất (Convergence in Probability: $X_n \xrightarrow{P} X$)

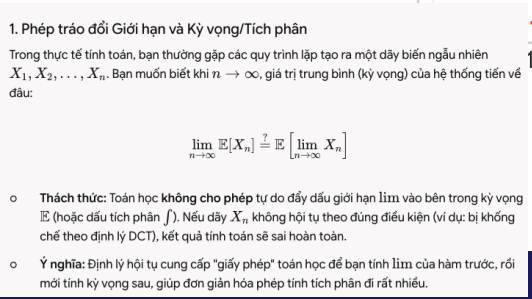
$$\forall \epsilon > 0, \quad \lim_{n\to\infty} P(|X_n - X| > \epsilon) = 0$$

Xác suất để xảy ra độ lệch $|X_n - X| > \epsilon$ sẽ triệt tiêu về 0 khi $n \to \infty$.

c. Hội tụ theo phân bố (Convergence in Distribution: $X_n \xrightarrow{\mathcal{D}} X$)

$$\lim_{n\to\infty} F_{X_n}(x) = F_X(x)$$

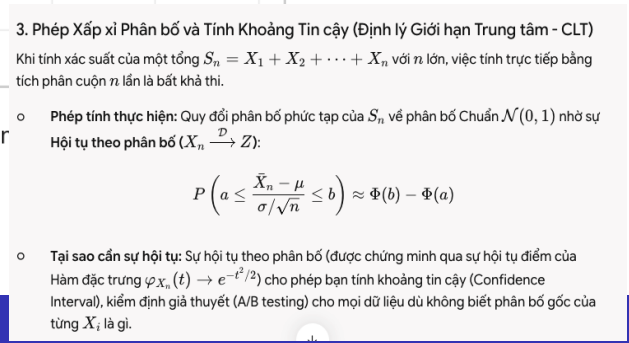
tại mọi điểm x mà hàm phân bố tích lũy $F_X(x)$ liên tục. Dạng này chỉ yêu cầu hình dạng phân bố xác suất của X_n tiến về phân bố của X, chứ không yêu cầu các giá trị $X_n(\omega)$ phải gần $X(\omega)$.


1. Phép tráo đổi Giới hạn và Kỳ vọng/Tích phân

Trong thực tế tính toán, bạn thường gặp các quy trình lặp tạo ra một dãy biến ngẫu nhiên $X_1, X_2, \ldots, X_n$. Bạn muốn biết khi $n \to \infty$, giá trị trung bình (kỳ vọng) của hệ thống tiến về đâu:

$$\lim_{n\to\infty} \mathbb{E}[X_n] \stackrel{?}{=} \mathbb{E}\left[\lim_{n\to\infty} X_n\right]$$

- **Thách thức:** Toán học **không cho phép** tự do đẩy dấu giới hạn $\lim$ vào bên trong kỳ vọng $\mathbb{E}$ (hoặc dấu tích phân $\int$). Nếu dãy X_n không hội tụ theo đúng điều kiện (ví dụ: bị khống chế theo định lý DCT), kết quả tính toán sẽ sai hoàn toàn.
- **Ý nghĩa:** Định lý hội tụ cung cấp "giấy phép" toán học để bạn tính $\lim$ của hàm trước, rồi mới tính kỳ vọng sau, giúp đơn giản hóa phép tính tích phân đi rất nhiều.


3. Phép Xấp xỉ Phân bố và Tính Khoảng Tin cậy (Định lý Giới hạn Trung tâm - CLT)

Khi tính xác suất của một tổng $S_n = X_1 + X_2 + \cdots + X_n$ với n lớn, việc tính trực tiếp bằng tích phân cuộn n lần là bất khả thi.

- **Phép tính thực hiện:** Quy đổi phân bố phức tạp của S_n về phân bố Chuẩn $\mathcal{N}(0,1)$ nhờ sự **Hội tụ theo phân bố** ($X_n \xrightarrow{\mathcal{D}} Z$):

$$P\left(a \le \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \le b\right) \approx \Phi(b) - \Phi(a)$$

- **Tại sao cần sự hội tụ:** Sự hội tụ theo phân bố (được chứng minh qua sự hội tụ điểm của Hàm đặc trưng $\varphi_{X_n}(t) \to e^{-t^2/2}$) cho phép bạn tính khoảng tin cậy (Confidence Interval), kiểm định giả thuyết (A/B testing) cho mọi dữ liệu dù không biết phân bố gốc của từng X_i là gì.

Convergence Almost Surely

Definition

$$X_n \xrightarrow{\text{a.s.}} X \quad \Longleftrightarrow \quad \mathbb{P}\Big(\lim_{n\to\infty} X_n = X\Big) = 1.$$

- Intuition: *pathwise* convergence for almost every outcome ω.
- Strongest of the three: implies in probability (hence in distribution).

1. "Quỹ đạo" (Sample Path) trong xác suất là gì?

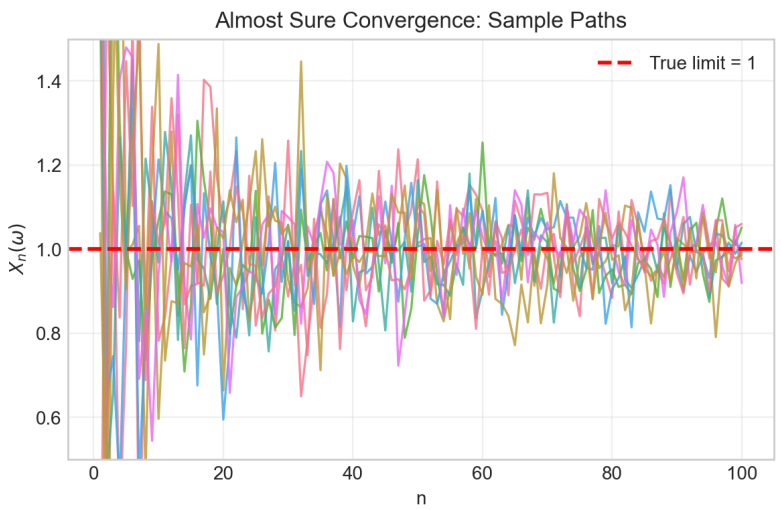
Về mặt toán học, mỗi biến ngẫu nhiên X_n không phải là một con số cố định, mà là một **hàm số** $X_n(\omega)$ xác định trên không gian mẫu Ω.

- Khi một phép thử ngẫu nhiên diễn ra, tự nhiên sẽ chọn ra một kết quả thực tế cụ thể $\omega \in \Omega$ (gọi là một kịch bản hay một "thế giới").
- Khi đã cố định kịch bản ω đó, dãy giá trị thu được theo bước $n = 1, 2, 3, \ldots$:

$$(X_1(\omega), X_2(\omega), X_3(\omega), \ldots, X_n(\omega), \ldots)$$

trở thành một **dãy số thực thông thường**.

Nếu bạn vẽ đồ thị của dãy số này theo trục thời gian/chỉ số n, bạn sẽ thu được một đường diễn biến gấp khúc. Đường diễn biến cá nhân này của kịch bản ω được gọi là **một quỹ đạo (sample path)**. PNG



2. Tại sao Hội tụ hầu chắc chắn ($a.s.$) lại xét "theo từng quỹ đạo"?

Định nghĩa của hội tụ hầu chắc chắn là:

$$\mathbb{P}\left(\left\{\omega \in \Omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\right\}\right) = 1$$

PNG

Điều này có nghĩa là ta đứng ở góc nhìn **dọc theo thời gian n của từng quỹ đạo cá nhân**:

1. Chọn ra một quỹ đạo ω.
2. Quan sát đường đi $X_n(\omega)$ khi $n \to \infty$.
3. Kiểm tra xem đường đi này có thực sự tiến dần và "lắng xuống" (settle) về giá trị $X(\omega)$ theo đúng nghĩa giới hạn số thực trong giải tích hay không. PNG

Khẳng định $\mathbb{P}(\ldots) = 1$ đồng nghĩa với việc: nếu bạn vẽ 1000 đường quỹ đạo ứng với 1000 kịch bản ω khác nhau, thì gần như **toàn bộ 1000 đường quỹ đạo đó đều sẽ hội tụ về đường** $X(\omega)$.

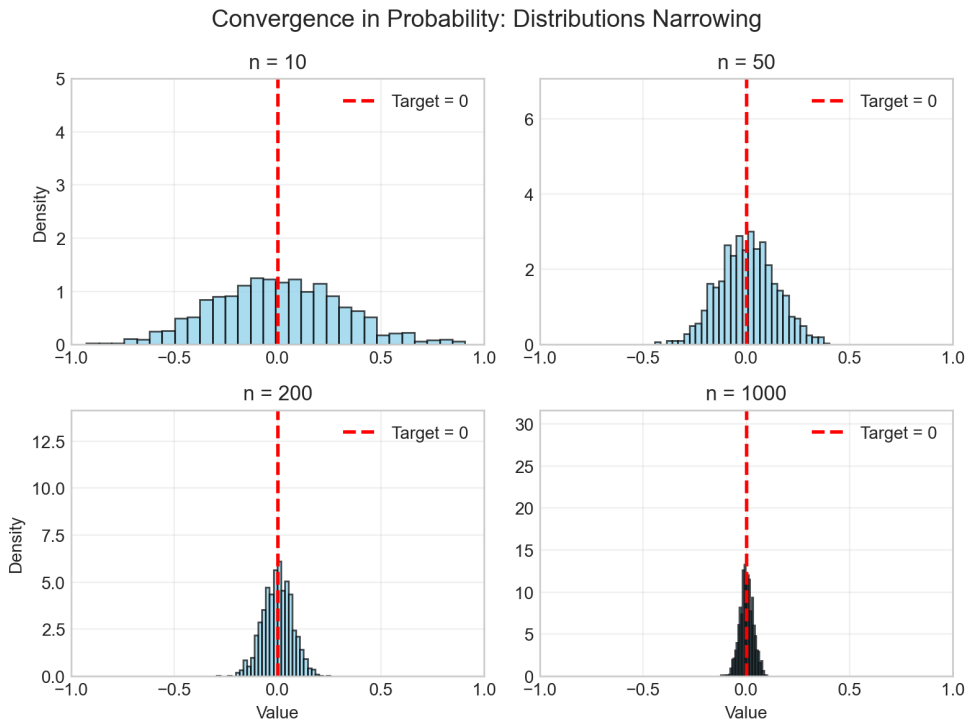
Example: $X_n = X + Y\,\mathbf{1}_A$ with $\mathbb{P}(A) = p$ small and fixed, and $Y \to 0$ deterministically.
Then $X_n \xrightarrow{\text{a.s.}} X$ when the pathwise perturbations vanish almost surely.

Convergence in Probability

Definition

$$X_n \xrightarrow{\mathbb{P}} X \quad \Longleftrightarrow \quad \forall\, \varepsilon > 0,\ \mathbb{P}(|X_n - X| > \varepsilon) \to 0.$$

- Intuition: large deviations become rare; individual paths may still oscillate.
- Weaker than a.s.; stronger than in distribution (in general settings).



Example: $X_n = \mathbf{1}_{\{U \le 1/n\}}$ with $U \sim \mathrm{Uniform}(0,1)$. Then $X_n \xrightarrow{\mathrm{P}} 0$ but $X_n \not\xrightarrow{\text{a.s.}} 0$.

Convergence in Distribution

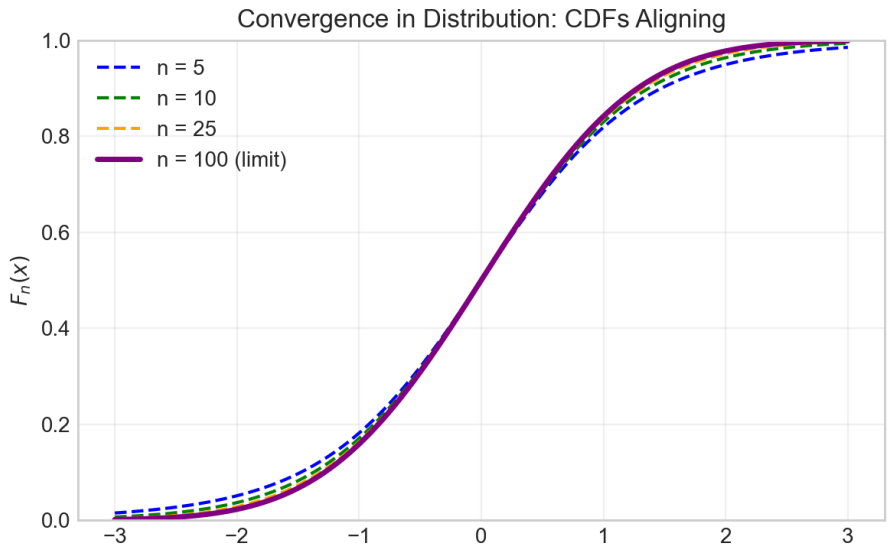
CDF-based definition

$$X_n \xrightarrow{d} X \quad \Longleftrightarrow \quad F_{X_n}(t) \to F_X(t) \text{ at all continuity points of } F_X.$$

Portmanteau (equivalent)

$$X_n \xrightarrow{d} X \quad \Longleftrightarrow \quad \mathbb{E}[f(X_n)] \to \mathbb{E}[f(X)] \quad \text{for all bounded, } \textit{continuous } f.$$

- Converging *shape* of the distribution; samples need not get close pointwise.
- Do not replace "continuous" by "measurable"; that would demand a stronger mode.



Summary & Intuition Map

Mode	Definition	Meaning	Visual	Implies
a.s.	$\mathbb{P}(\lim X_n = X) = 1$	Pathwise for a.e. outcome	paths settle	in P
in P	$\mathbb{P}(\lvert X_n - X\rvert > \varepsilon) \to 0$	Large deviations rare	histograms narrow	in $\mathcal{D}$
in $\mathcal{D}$	$F_{X_n} \to F_X$ at cont. pts	Laws converge	CDFs align	—

Key Hierarchy

Almost surely $\Rightarrow$ In probability $\Rightarrow$ In distribution

Takeaway

Stronger modes give more pathwise closeness; weaker modes only guarantee distribution-level agreement.

Outline

1. Modeling Motivation
2. Foundations and Random Variables
3. Statistics and Convergence
4. Limit Theorems
5. Exercises
6. Summary and References

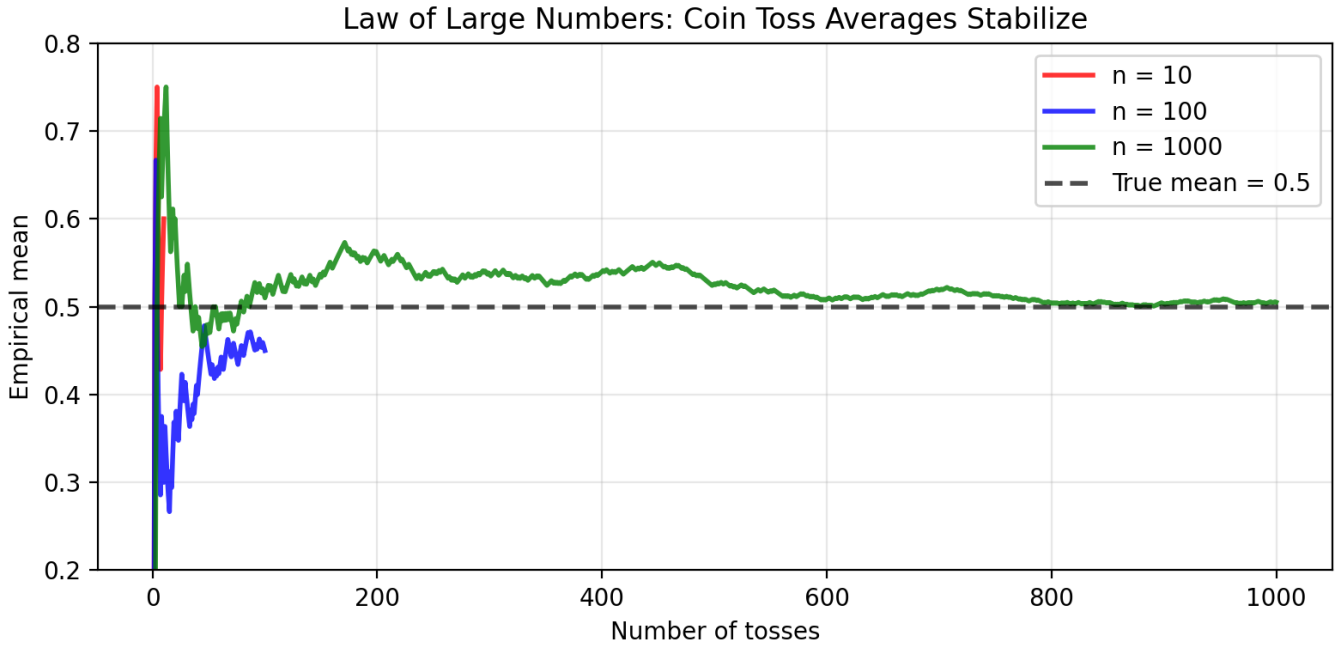
Why do averages stabilize?

Individual outcomes are random and fluctuate:

- A single coin toss: completely unpredictable (heads or tails).
- But the average of many tosses becomes remarkably predictable.

Key insight:

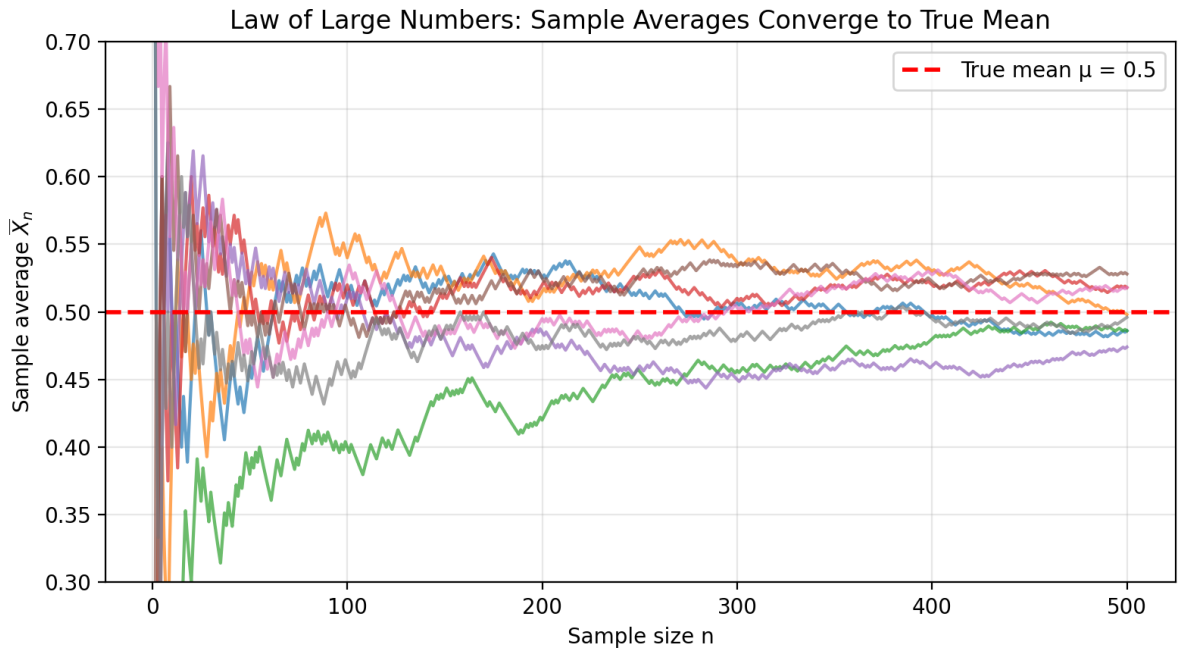
- Averaging across many samples cancels out randomness.
- The average becomes more predictable as sample size grows.
- This stabilization is guaranteed by mathematical laws.



Empirical averages get closer to the true mean

Multiple sample paths of Bernoulli(0.5):

- Early averages fluctuate wildly between different sample paths.
- As n increases, all paths converge toward the true mean $\mu = 0.5$.
- Individual variation decreases; central tendency emerges.



Key message

"The law of large numbers tells us this stabilization is guaranteed."

Law of Large Numbers (Informal)

Informal statement

"When we take more and more independent observations, the sample average converges to the true mean."

In everyday language:

- "Data averages out."
- Random fluctuations cancel when you have enough data.
- Larger samples $\Rightarrow$ more reliable estimates.

Why this matters for statistics:

- Justifies using sample means to estimate population parameters.
- Explains why polling works (with sufficient sample size).
- Foundation for Monte Carlo methods and simulation.

Law of Large Numbers (Formal)

Formal statement (Weak LLN)

Let $X_1, X_2, \ldots$ be i.i.d. random variables with expectation $\mu = \mathbb{E}[X_1]$.

Define the sample mean: $\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$.

Then: $\overline{X}_n \xrightarrow{\mathbb{P}} \mu$.

Translation to plain language:

- For any $\varepsilon > 0$: $\mathbb{P}(|\overline{X}_n - \mu| > \varepsilon) \to 0$ as $n \to \infty$.
- The probability of large deviations from the true mean vanishes.
- Sample averages concentrate around the population mean.

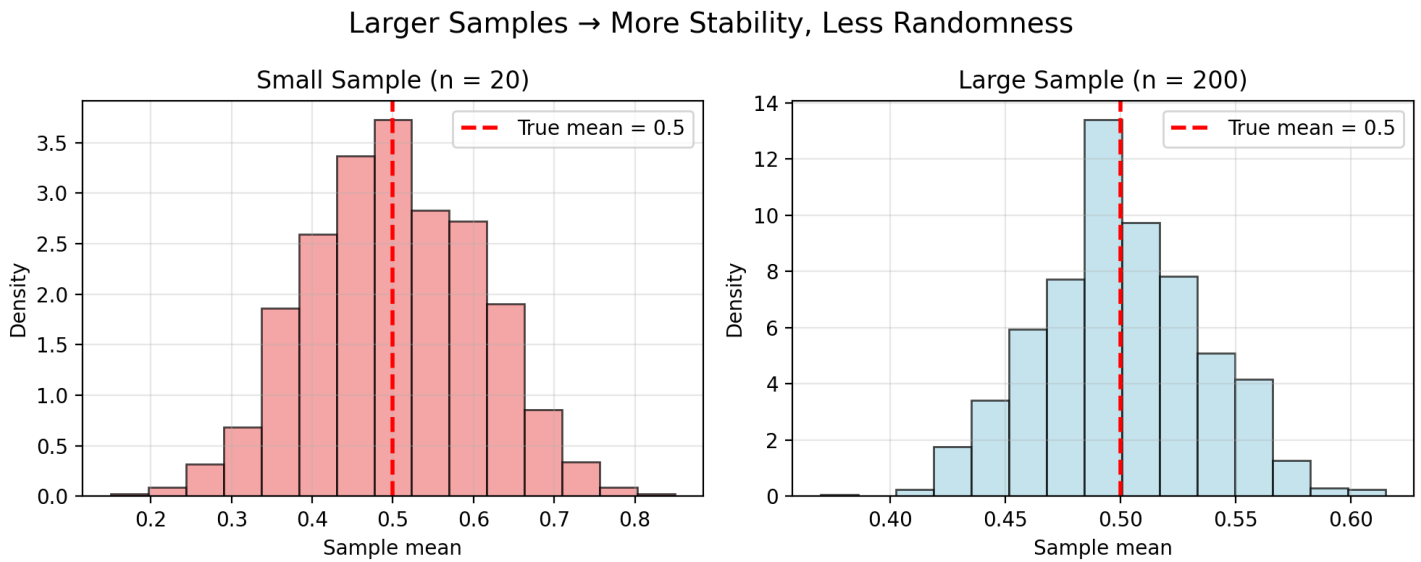
Why LLN Matters for Applied Statistics

Core justification for statistical inference:

- Sample averages (and estimators) approximate population averages.
- Foundation for estimation theory.
- More data $\Rightarrow$ more reliable results.

Key principles:

- Larger samples $\Rightarrow$ more stability, less randomness.
- Random quantities become "deterministic-looking" with enough data.
- Basis for confidence intervals, hypothesis tests, and prediction.

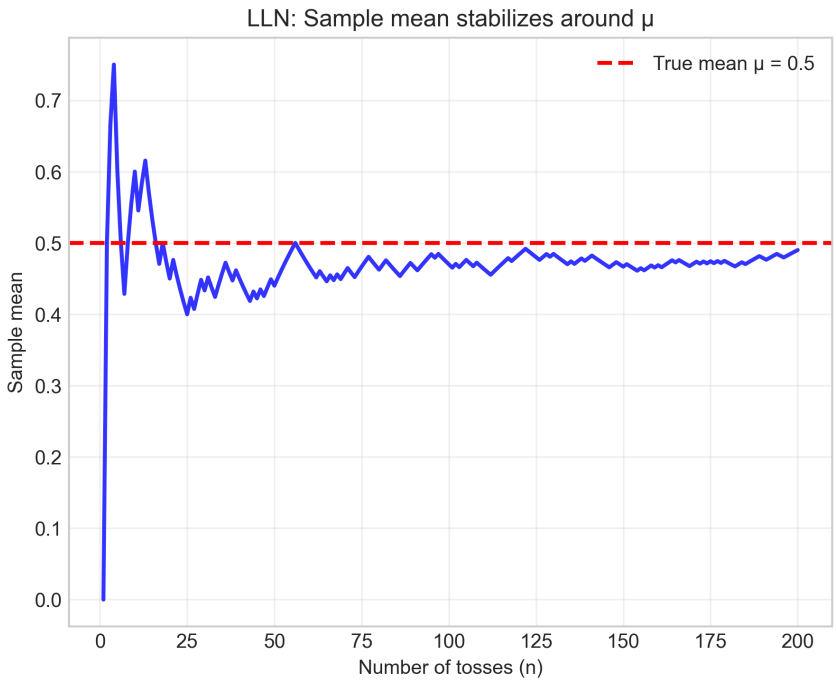


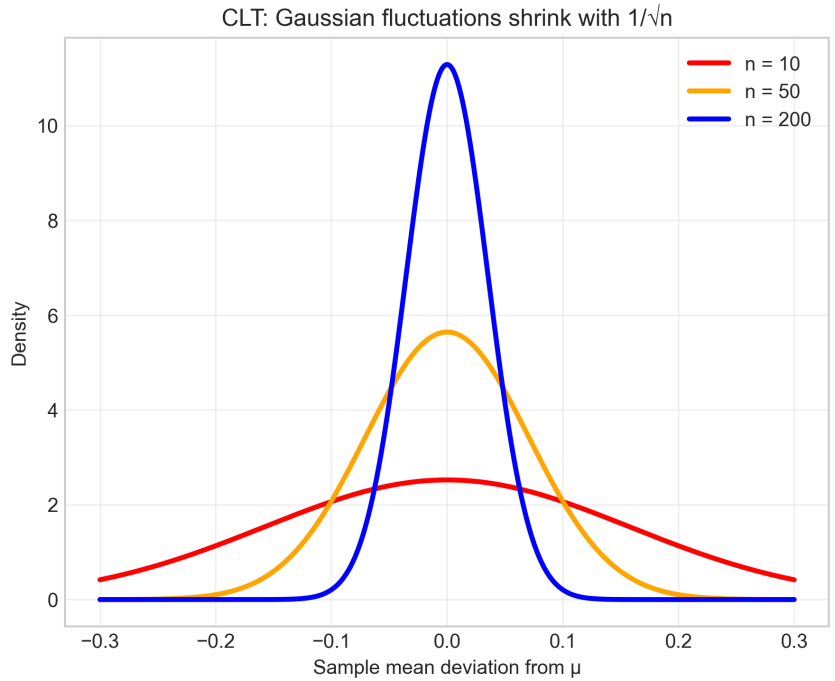
Small sample vs large sample: histogram comparison showing reduced variability.

From stabilization to distributional shape

Motivation & Intuition

- LLN: sample mean $\overline{X}_n$ converges to the true mean μ.
- CLT: answers *how fast* (scaling by $\sqrt{n}$) and *with what shape* (Gaussian).





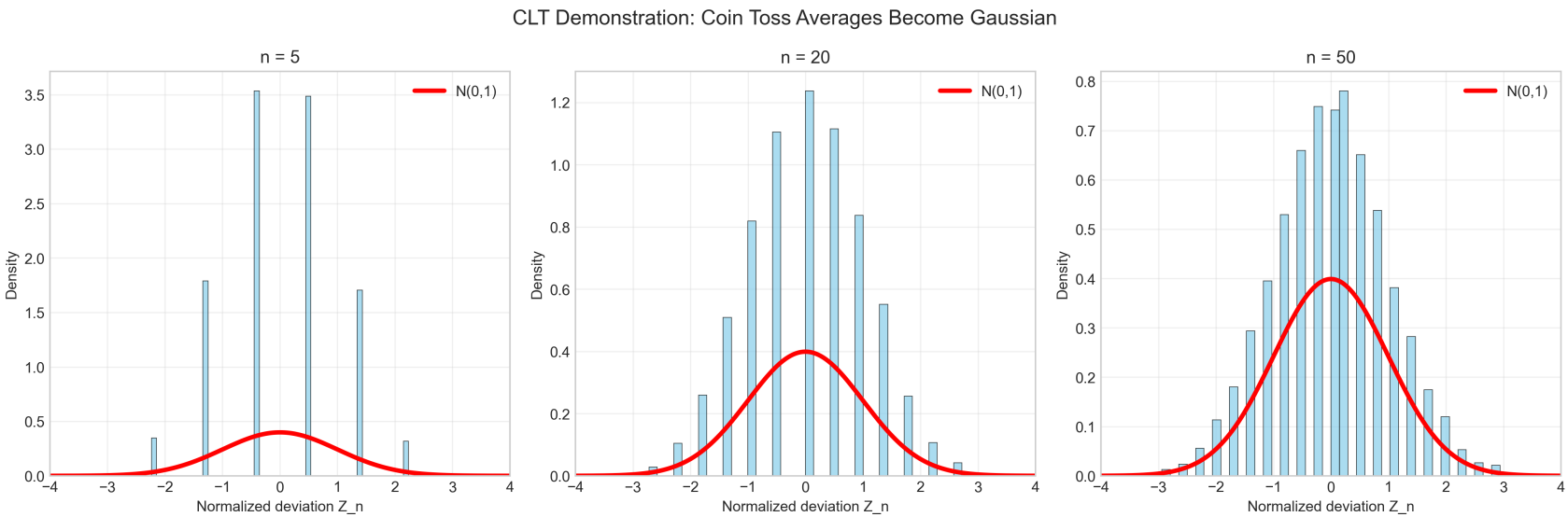
Key insight: LLN = convergence to mean. CLT = Gaussian fluctuations shrinking with $1/\sqrt{n}$.

Coin toss averages become Gaussian

Visual Demonstration

Experiment: Toss a fair coin (0/1), repeat many times.

- Simulate 10,000 repetitions of sample means for $n = 5, 20, 50$.
- Compute normalized deviations: $Z_n = \sqrt{n}(\overline{X}_n - 0.5)$.



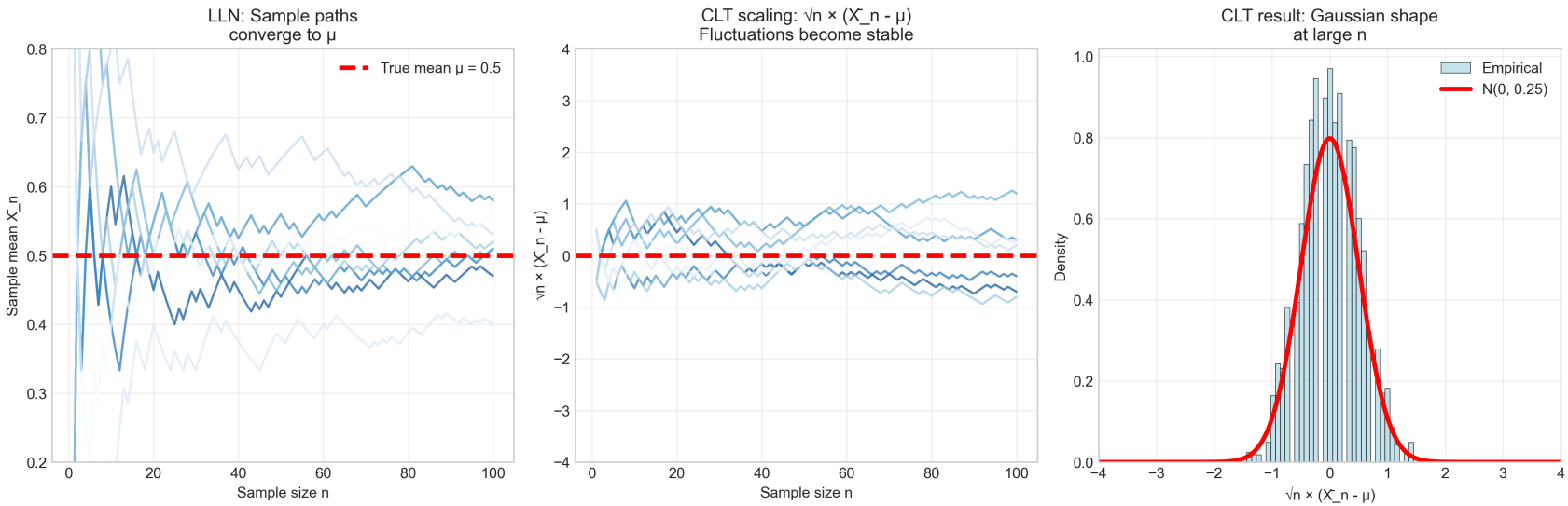
As n grows, Z_n looks increasingly Gaussian.

Central Limit Theorem (Informal)

Informal Statement

"LLN says averages stabilize. CLT says that if we zoom in at the scale $\sqrt{n}$, fluctuations are Gaussian."

Key insight: CLT provides both *speed* and *shape* of convergence.



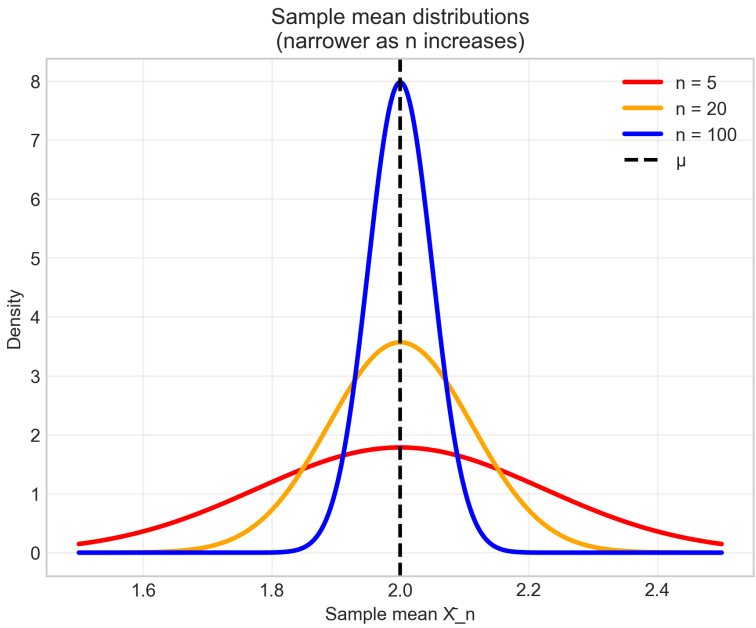
Left: Multiple sample paths converge to μ. **Middle:** After $\sqrt{n}$-scaling, fluctuations stabilize. **Right:** At large n, the scaled deviations are Gaussian.

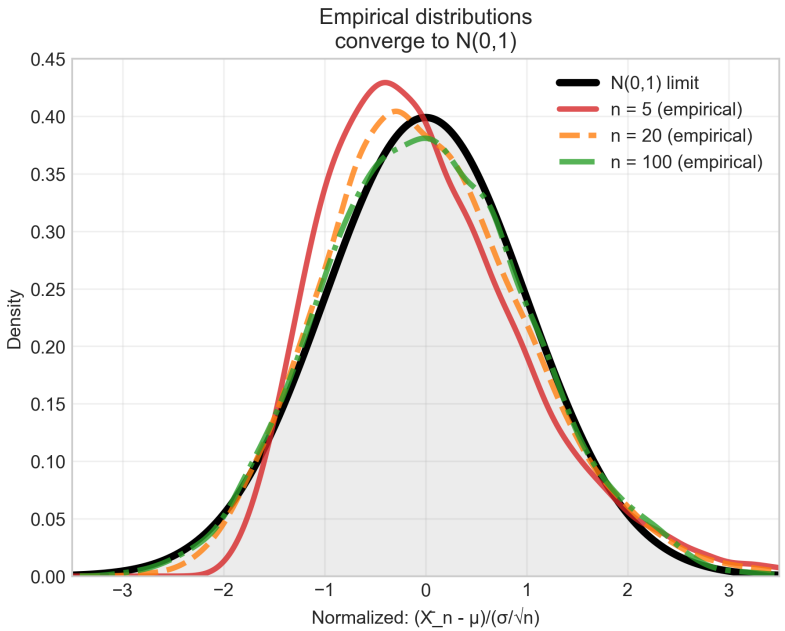
Central Limit Theorem (Formal)

Formal Statement

Let $X_1, X_2, \ldots$ i.i.d. with mean μ, variance $\sigma^2 < \infty$.
Sample mean: $\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$.
Then: $\sqrt{n}\,\frac{\overline{X}_n - \mu}{\sigma} \xrightarrow{d} \mathcal{N}(0,1)$.

Interpretation: Fluctuations shrink at rate $1/\sqrt{n}$ and are Gaussian-shaped.

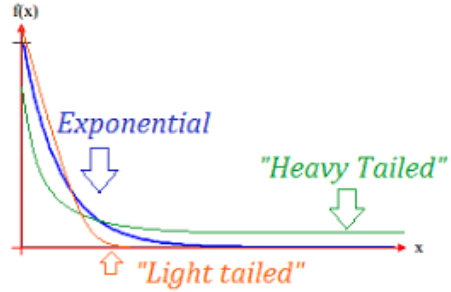




Why the CLT matters for Applied Statistics

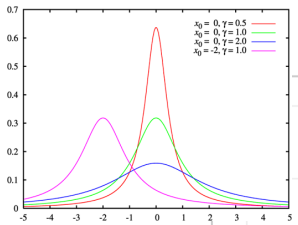
Định lý Giới hạn Trung tâm (CLT) chỉ có hiệu lực khi phân bố ban đầu thỏa mãn một điều kiện kiên quyết: **Phương sai phải hữu hạn** ($\sigma^2 < \infty$) (và kỳ vọng hữu hạn $\mu < \infty$).

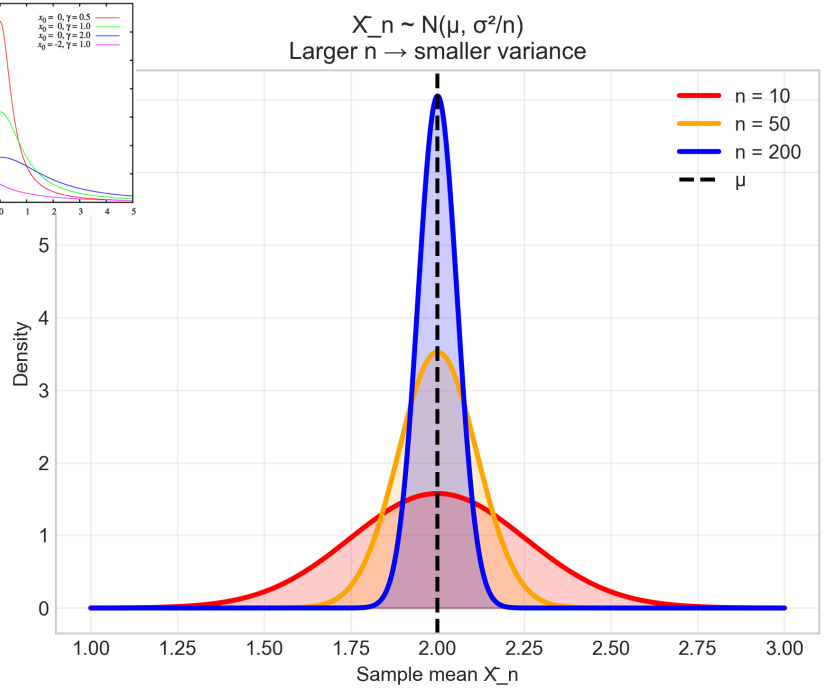
Dưới đây là các trường hợp ngoại lệ kinh điển mà CLT hoàn toàn thất bại:

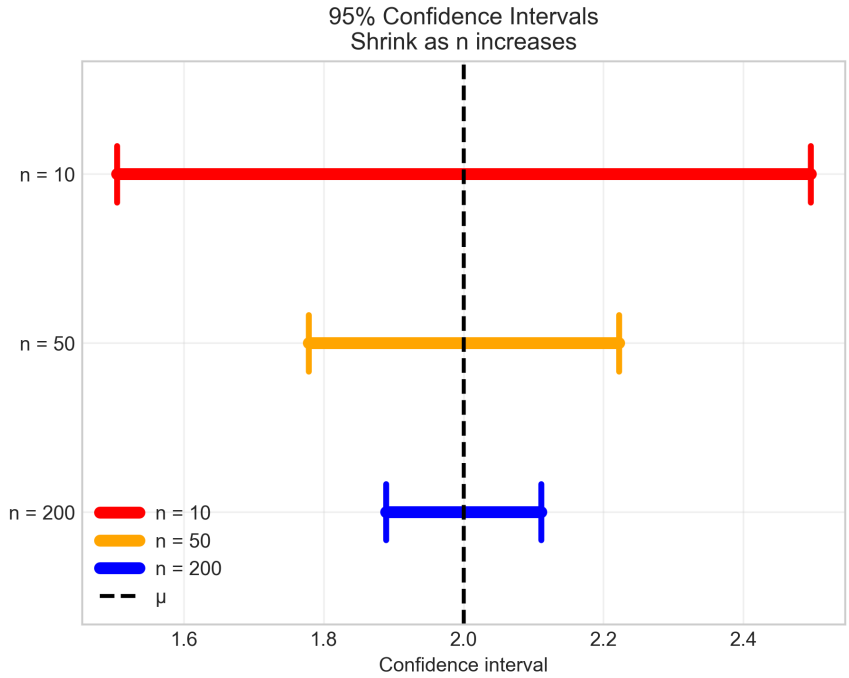


Practical Consequences

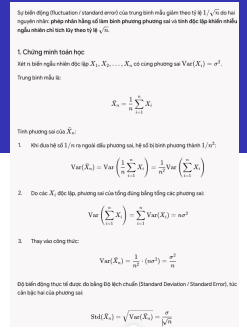
- Approximation: $\overline{X}_n \approx \mathcal{N}(\mu, \sigma^2/n)$.
- Larger $n \Rightarrow$ smaller variance $\Rightarrow$ narrower confidenc
- CLT underpins confidence intervals and hypothesis tests.





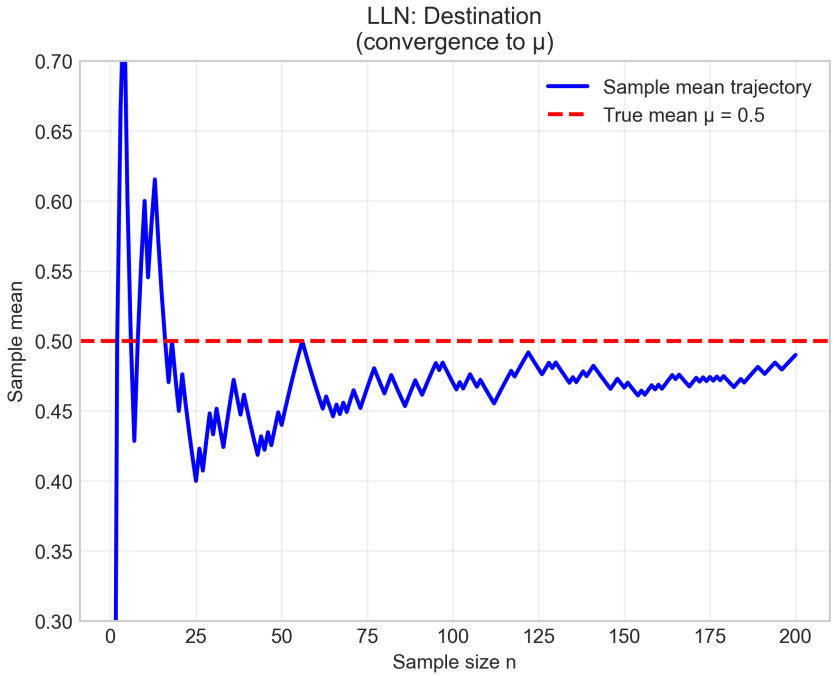


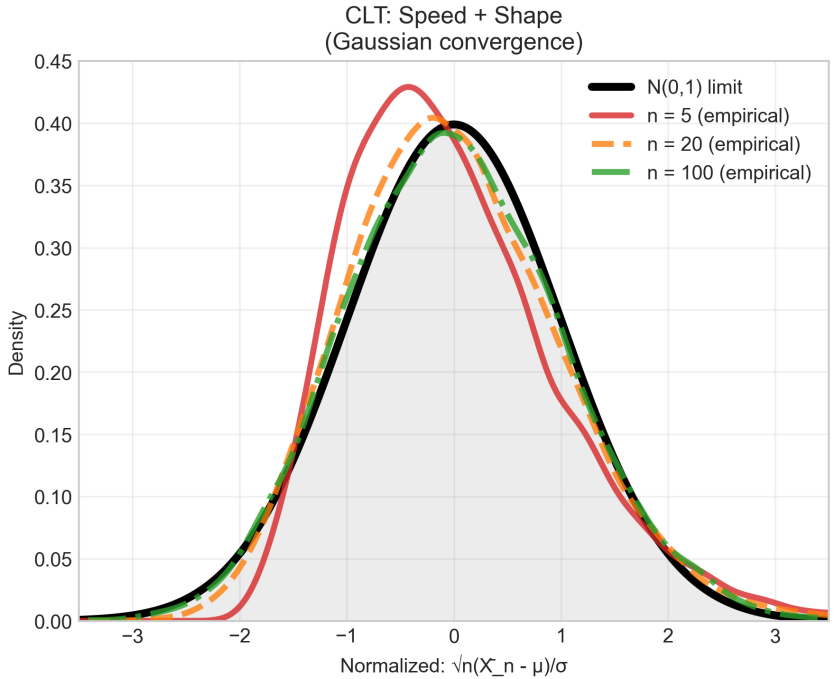
Key Insights from the CLT



Takeaways

- LLN: averages converge to μ.
- CLT: fluctuations shrink at rate $1/\sqrt{n}$ and are Gaussian-shaped.
- This dual perspective makes inference possible.





Remember: LLN = destination. CLT = speed + shape of convergence.

Outline

1. Modeling Motivation
2. Foundations and Random Variables
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Exercises (Theory)

3. Hệ số bất đối xứng (Skewness) của $X \sim \text{Bernoulli}(p)$

Hàm xác suất: $\mathbb{P}(X=1)=p, \mathbb{P}(X=0)=1-p$.

- Kỳ vọng và độ lệch chuẩn:

$$\mu = \mathbb{E}[X] = p$$

$$\sigma^2 = \text{Var}(X) = p(1-p) \implies \sigma = \sqrt{p(1-p)}$$

- Tính Moment trung tâm cấp 3 ($\mu_3 = \mathbb{E}[(X-p)^3]$):

$$\mathbb{E}[(X-p)^3] = p(1-p)^3 + (1-p)(0-p)^3 = p(1-p)\left[(1-p)^2 - p^2\right]$$

$$= p(1-p)(1-2p+p^2-p^2) = p(1-p)(1-2p)$$

- Hệ số Skewness γ_1:

$$\gamma_1 = \frac{\mathbb{E}[(X-\mu)^3]}{\sigma^3} = \frac{p(1-p)(1-2p)}{(p(1-p))^{3/2}} = \frac{1-2p}{\sqrt{p(1-p)}} \quad \text{(đpcm)}$$

1. $X \sim \text{Uniform}(0,1)$

- Kỳ vọng $\mathbb{E}[X]$:

$$\mathbb{E}[X] = \int_0^1 x \cdot 1\, dx = \left[\frac{x^2}{2}\right]_0^1 = \frac{1}{2}$$

- Phương sai $\text{Var}(X)$:

$$\mathbb{E}[X^2] = \int_0^1 x^2\, dx = \left[\frac{x^3}{3}\right]_0^1 = \frac{1}{3}$$

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{1}{3} - \left(\frac{1}{2}\right)^2 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$$

- Trung vị (Median):
  Giải phương trình $F(m) = P(X \le m) = 0.5 \implies m = 0.5$.
- Phân vị cấp 0.9 ($q_{0.9}$):
  Giải phương trình $F(q_{0.9}) = P(X \le q_{0.9}) = 0.9 \implies q_{0.9} = 0.9$.

1. If $X \sim \text{Uniform}(0,1)$, compute $\mathbb{E}[X]$, $\text{Var}(X)$, median, and $q_{0.9}$.
2. If $X \sim \text{Poisson}(3)$, compute $\mathbb{P}(X \le 1)$ and $\mathbb{E}[X(X-1)]$.
3. Show that for $X \sim \text{Bernoulli}(p)$, skewness $\gamma_1 = \dfrac{1-2p}{\sqrt{p(1-p)}}$.
4. Prove WLLN using Chebyshev for i.i.d. with finite variance.

maclaurin

2. $X \sim \text{Poisson}(3)$ $(\lambda = 3)$

- Tính $\mathbb{P}(X \le 1)$:

$$\mathbb{P}(X \le 1) = \mathbb{P}(X=0) + \mathbb{P}(X=1) = \frac{e^{-3}3^0}{0!} + \frac{e^{-3}3^1}{1!} = e^{-3}(1+3) = 4e^{-3} \approx 0.$$

- Tính $\mathbb{E}[X(X-1)]$:
  Sử dụng moment thừa số (factorial moment) của phân bố Poisson:

$$\mathbb{E}[X(X-1)] = \sum_{k=0}^{\infty} k(k-1)\frac{e^{-\lambda}\lambda^k}{k!} = \sum_{k=2}^{\infty} \frac{e^{-\lambda}\lambda^k}{(k-2)!} = \lambda^2 e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!} = \lambda^2 e^{-\lambda} \cdot e^{\lambda}$$

Với $\lambda = 3$:

$$\mathbb{E}[X(X-1)] = 3^2 = 9$$

4. Chứng minh Luật Yếu Số Lớn (WLLN) dùng Bất đẳng thức Chebyshev

Cho $X_1, X_2, \ldots, X_n$ i.i.d. có $\mathbb{E}[X_i] = \mu$ và $\text{Var}(X_i) = \sigma^2 < \infty$. Đặt $\bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i$.

1. Tính các đặc trưng của $\bar{X}_n$:

$$\mathbb{E}[\bar{X}_n] = \mu, \quad \text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}$$

2. Áp dụng Bất đẳng thức Chebyshev cho $\bar{X}_n$:
   Với mọi $\epsilon > 0$:

$$\mathbb{P}(|\bar{X}_n - \mu| \ge \epsilon) \le \frac{\text{Var}(\bar{X}_n)}{\epsilon^2} = \frac{\sigma^2}{n\epsilon^2}$$

3. Lấy giới hạn khi $n \to \infty$:
   Vì $\lim_{n\to\infty} \frac{\sigma^2}{n\epsilon^2} = 0$, theo Nguyên lý kẹp:

$$\lim_{n\to\infty} \mathbb{P}(|\bar{X}_n - \mu| \ge \epsilon) = 0 \quad \text{(đpcm)}$$

Practical Preview

- EDA on heights dataset
- CLT simulation; Poisson fit to defects

Reference

See practical tasks in the lesson materials and starter code in the repository.

Outline

1. Modeling Motivation
2. Foundations and Random Variables
3. Statistics and Convergence
4. Limit Theorems
5. Exercises
6. Summary and References

Summary

- Bridge descriptive to probabilistic modeling

References

- Casella and Berger, Statistical Inference
- Wasserman, All of Statistics
- Grimmett and Stirzaker, Probability and Random Processes